MIL-M-28787/81A(EC)
29 June 1979
SUPERSEDING
MIL-M-28787/81(EC)
8 December 1976

MILITARY SPECIFICATION

*MODULES, STANDARD ELECTRONIC

*MULTIPLEXER, DIGITAL, KEY CODE KHL

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic multiplexer module.

1.2 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table II.

Name	Multiplexer
Family	Digital
Key code	KHL
Size	1A
Weight	35 grams maximum
Failure rate	0.752 failures/10^6 hours maximum
Environment	Class I as specified in MIL-STD-1389
Maximum power dissipation	1,155 mW (outputs not loaded)
Supply voltage	5 Vdc ±10 percent
Input voltage	
Low level	-0.5 V minimum, 0.8 V maximum
High level	2.0 V minimum, V_{CC} maximum
Input current, D0-D7, 3A, 3B, 3C, STROBE 3	
Low level (V_{IL} = 0.0 V)	-2.2 mA maximum
(V_{IL} = 0.5 V)	-2.0 mA maximum
High level (V_{IH} = 2.5 V)	50 µA maximum
(V_{IH} = 5.5 V)	1.0 mA maximum

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, ATTN: ELEX 5043, Department of the Navy, Washington, D.C. 20360 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:59Z
Check the source to verify that this is the current version before use.

Input current, 1-2A, 1-2B, 1-2C, STROBE 1-2	
Low level (V_{IL} = 0.0 V)	-4.4 mA maximum
(V_{IL} = 0.5 V)	-4.0 mA maximum
High level (V_{IH} = 2.5 V)	100 µA maximum
(V_{IH} = 5.5 V)	2.0 mA maximum
Output voltage	
Low level (I_{OL} = 20 mA)	0.5 V maximum
High level (I_{OH} = -1.0 mA)	2.5 V maximum
Output drive current maximum capability	
Low level (V_{OL} = 0.5 V)	20 mA
High level (V_{OH} = 2.5 V)	-1.0 mA
Propagation delay time	
t_{PLH} - D0-D7 to W output	
C_L = 50 pF	2.0 ns minimum, 9.5 ns maximum
C_L = 150 pF	4.0 ns minimum, 11.5 ns maximum
t_{PLH} - D0-D7 to Y output	
C_L = 50 pF	2.0 ns minimum, 15.0 ns maximum
C_L = 150 pF	4.0 ns minimum, 17.0 ns maximum
t_{PLH} - A, B, C, to W output	
C_L = 50 pF	2.0 ns minimum, 18.0 ns maximum
C_L = 150 pF	4.0 ns minimum, 20.0 ns maximum
t_{PLH} - A, B, C, to Y output	
C_L = 50 pF	2.0 ns minimum, 25.0 ns maximum
C_L = 150 pF	4.0 ns minimum, 27.0 ns maximum
t_{PLH} - STROBE to W output	
C_L = 50 pF	2.0 ns minimum, 15.0 ns maximum
C_L = 150 pF	4.0 ns minimum, 17.0 ns maximum
t_{PLH} - STROBE to Y output	
C_L = 50 pF	2.0 ns minimum, 22.0 ns maximum
C_L = 150 pF	4.0 ns minimum, 24.0 ns maximum
t_{PHL} - D0-D7 to W output	
C_L = 50 pF	2.0 ns minimum, 9.5 ns maximum
C_L = 150 pF	4.0 ns minimum, 11.5 ns maximum
t_{PHL} - D0-D7 to Y output	
C_L = 50 pF	2.0 ns minimum, 15.0 ns maximum
C_L = 150 pF	4.0 ns minimum, 17.0 ns maximum
t_{PHL} - A, B, C, to W output	
C_L = 50 pF	2.0 ns minimum, 18.0 ns maximum
C_L = 150 pF	4.0 ns minimum, 20.0 ns maximum
t_{PHL} - A, B, C, to Y output	
C_L = 50 pF	2.0 ns minimum, 25.0 ns maximum
C_L = 150 pF	4.0 ns minimum, 27.0 ns maximum

Source: https://assist.dla.mil -- Downloaded: 20260-20T15:59Z
Check the source to verify that this is the current version before use.

Propagation delay time - Continued

t_{PHL} - STROBE to W output

C_L = 50 pF	2.0 ns minimum, 15.0 ns maximum
C_L = 150 pF	4.0 ns minimum, 17.0 ns maximum

t_{PHL} - STROBE to Y output

C_L = 50 pF	2.0 ns minimum, 22.0 ns maximum
C_L = 150 pF	4.0 ns minimum, 24.0 ns maximum

1.3 Module function. This module contains three programmable eight-input multiplexers. Each multiplexer receives eight input data lines (D0 through D7), three input data select lines (A, B, and C), and an input-enable line (STROBE). The input data line used is selected by the state of the data select inputs. This data is transferred to the output (Y) and its complement (W) when the input enable (STROBE) is active (low). When it is not activated, the output is low regardless of all other inputs. Multiplexers 1 and 2 have a common input enable (STROBE). Multiplexer 3 has a separate input enable (STROBE) Multiplexers 1 and 2 have common data select lines (A, B, and C). Multiplexer 3 has a separate data select lines (A, B, and C).

1.3.1 Input to output relationship. Each multiplexer has the input to output relationship of table I for positive logic.

1.3.2 Parallel outputs. Paralleling of outputs is not permissible.

2 APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE I. Input to output relationship.

Input											Output	
Data select			Data								Enable		
C	B	A	D0	D1	D2	D3	D4	D5	D6	D7	STROBE	Y	W
X	X	X	X	X	X	X	X	X	X	X	H	L	H
L	L	L	L	X	X	X	X	X	X	X	L	L	H
L	L	L	H	X	X	X	X	X	X	X	L	H	L
L	L	H	X	L	X	X	X	X	X	X	L	L	H
L	L	H	X	H	X	X	X	X	X	X	L	H	L
L	H	L	X	X	L	X	X	X	X	X	L	L	H
L	H	L	X	X	H	X	X	X	X	X	L	H	L
L	H	H	X	X	X	L	X	X	X	X	L	L	H
L	H	H	X	X	X	H	X	X	X	X	L	H	L
H	L	L	X	X	X	X	L	X	X	X	L	L	H
H	L	L	X	X	X	X	H	X	X	X	L	H	L
H	L	H	X	X	X	X	X	L	X	X	L	L	H
H	L	H	X	X	X	X	X	H	X	X	L	H	L
H	H	L	X	X	X	X	X	X	L	X	L	L	H
H	H	L	X	X	X	X	X	X	H	X	L	H	L
H	H	H	X	X	X	X	X	X	X	L	L	L	H
H	H	H	X	X	X	X	X	X	X	H	L	H	L

NOTES
1. H = High-level voltage.
2. L = Low-level voltage.
3. X = Don't care condition.

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:59Z
Check the source to verify that this is the currentversion before use.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.752 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table II and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be KHL.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/81-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The logic diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with notes 1 and 2 of table II.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table II.

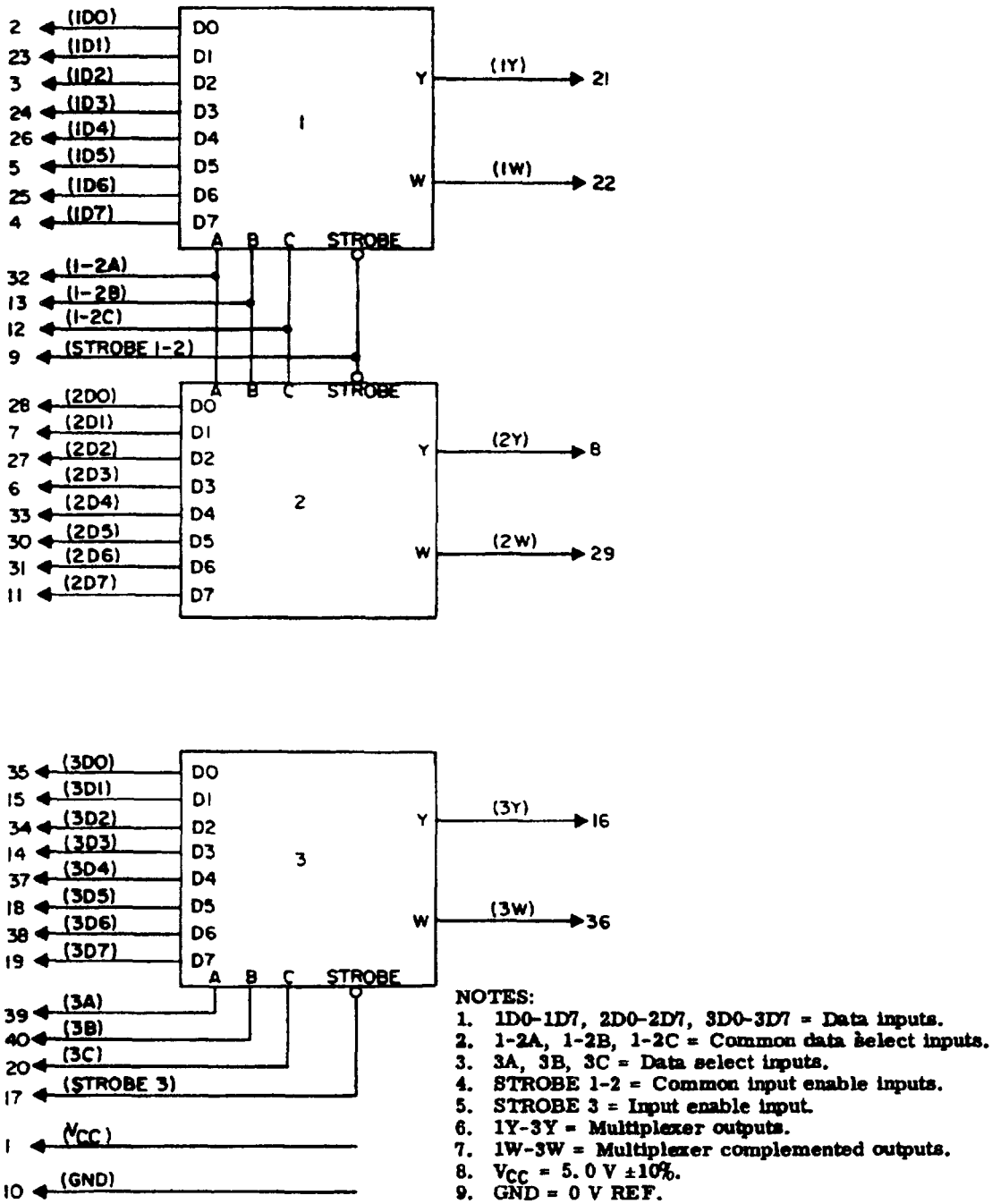
3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table II for a power supply voltage (V_{CC}) of 5.0 $\pm$0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc $\pm$1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:59Z
Check the source to verify that this is the current version before use.



NOTES:
1. 1D0-1D7, 2D0-2D7, 3D0-3D7 = Data inputs.
2. 1-2A, 1-2B, 1-2C = Common data select inputs.
3. 3A, 3B, 3C = Data select inputs.
4. STROBE 1-2 = Common input enable inputs.
5. STROBE 3 = Input enable input.
6. 1Y-3Y = Multiplexer outputs.
7. 1W-3W = Multiplexer complemented outputs.
8. V_{CC} = 5.0 V ±10%.
9. GND = 0 V REF.

FIGURE 1. KHL module function diagram, digital triple 8-input multiplexer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE II. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation	III	S	10	-	1.0	-	10	-	1.0	-	MΩ
B1-B6	V_{OUT} (low-loaded)	IV	S	0.01	1.00	0.01	1.00	0.01	1.00	0.01	1.00	V
C1-C28	Input breakdown current, D0-D7, 3A, 3B, 3C,STROBE 3	V	S	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	mA
C29-C32	Input breakdown current, 1-2A, 1-2B, 1-2C,STROBE 1-2	V	S	0.0	2.0	0.0	2.0	0.0	2.0	0.0	2.0	mA
D1-D6	Output leakage current	VI	X	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	mA
E1-E6	Output short circuit current	VII	X	-40	-100	-40	-100	-40	-100	-40	-100	mA
F1-F25	Functional test	VIII	X	Test vectors must produce output vectors as specified in test table								
G1-G32	Input clamp diode voltage	IX	X	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	V
H1-H6	Static output voltage, low level	X	X	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	V
J1-J6	Static output voltage, high level	XI	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
K1-K28	Input current, high level, D0-D7 3A, 3B, 3C,STROBE 3	XII	X	0	50	0	50	0	50	0	50	µA
K29-K32	Input current, high level, 1-2A, 1-2B, 1-2C,STROBE 1-2	XII	X	0	100	0	100	0	100	0	100	µA
L1-L28	Input current, low level, D0-D7, 3A, 3B, 3C, STROBE 3	XIII	X	-0.1	-2.2	-0.1	-2.2	-0.1	-2.2	-0.1	-2.2	mA
L29-L32	Input current, low level, 1-2A, 1-2B, 1-2C, STROBE 1-2	XIII	X	-0.2	-4.4	-0.2	-4.4	-0.2	-4.4	-0.2	-4.4	mA
M	Power supply current	XIV	X	45	210	45	210	45	210	45	210	mA

See notes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026/0-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE II. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
N1,N3,N5	Transition time (t_{TLH}), C_L = 50 pF	XV	S	2.0	15.0	2.0	15.0	2.0	18.0	2.0	18.0	ns
N2,N4,N6	Transition time (t_{TLH}), C_L = 50 pF	XV	S	2.0	12.0	2.0	12.0	2.0	15.0	2.0	15.0	ns
N7,N9,N11	Transition time (t_{TLH}), C_L = 150 pF	XV	Q	2.0	27.0	2.0	27.0	2.0	30.0	2.0	30.0	ns
N8,N10,N12	Transition time (t_{TLH}), C_L = 150 pF	XV	Q	2.0	24.0	2.0	24.0	2.0	27.0	2.0	27.0	ns
P1-P6	Transition time (t_{THL}), C_L = 50 pF	XV	S	2.0	6.0	2.0	6.0	2.0	8.0	2.0	8.0	ns
P7-P12	Transition time (t_{THL}), C_L = 150 pF	XV	Q	2.0	11.0	2.0	11.0	2.0	13.0	2.0	13.0	ns
Q1-Q15	Propagation delay time (t_{PLH}), D0-D4 to W output, C_L = 50 pF	XV	S	2.0	8.5	2.0	8.5	2.0	9.5	2. 0	9.5	ns
Q16-Q21	Propagation delay time (t_{PLH}), D1,D5 to Y output, C_L = 50 pF	XV	S	2.0	13.5	2.0	13.5	2.0	15.0	2.0	15.0	ns
Q22-Q30	Propagation delay time (t_{PLH}), A,B,Cto W output, C_L = 50 pF	XV	S	2.0	16.5	2.0	16.5	2.0	18.0	2.0	18.0	ns
Q31-Q33	Propagation delay time (t_{PLH}),1-2A, 1-2B,3C toY output, C_L = 50 pF	XV	S	2.0	23.0	2.0	23.0	2.0	25.0	2.0	25.0	ns
Q34-Q36	Propagation delay time (t_{PLH}), STROBE to W output, C_L = 50 pF	XV	S	2.0	14.0	2.0	14.0	2.0	15.0	2.0	15.0	ns
Q37-Q39	Propagation delay time (t_{PLH}), STROBE to Y output, C_L = 50 pF	XV	S	2.0	20.5	2.0	20.5	2.0	22.0	2.0	22.0	ns
Q40-Q54	Propagation delay time (t_{PLH}), D0-D4 to W output, C_L = 150 pF	XV	Q	4.0	10.5	4.0	10.5	4.0	11.5	4.0	11.5	ns

See notes at end of table.

Source: https://assist.dla.mil -- Downloaded: 20260-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE II. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
Q55-Q60	Propagation delay time (t_{PLH}), D1, D5 to Y output, C_L = 150 pF	XV	Q	4.0	16.5	4.0	16.5	4.0	18.0	4.0	18.0	ns
Q61-Q69	Propagation delay time (t_{PLH}), A, B, C to W output, C_L = 150 pF	XV	Q	4.0	18.0	4.0	18.0	4.0	20.0	4.0	20.0	ns
Q70-Q72	Propagation delay time (t_{PLH}),1-2A 1-2B, 3C to Y output, C_L = 150 pF	XV	Q	4.0	25.0	4.0	25.0	4.0	27.0	4.0	27.0	ns
Q73-Q75	Propagation delay time (t_{PLH}), STROBE to W output, C_L = 150 pF	XV	Q	4.0	16.0	4.0	16.0	4.0	17.0	4.0	17.0	ns
Q76-Q78	Propagation delay time (t_{PLH}), STROBE to Y output, C_L =150 pF	XV	Q	4.0	22.5	4.0	22.5	4.0	24.0	4.0	24.0	ns
R1-R15	Propagation delay time (t_{PHL}), D0, D4-D7 to W output, C_L = 50 pF	XV	S	2.0	8.5	2.0	8.5	2.0	9.5	2.0	9.5	ns
R16-R21	Propagation delay time (t_{PHL}), D1, D5 to Y output, C_L = 50 pF	XV	S	2.0	13.5	2.0	13.5	2.0	15.0	2.0	15.0	ns
R22-R30	Propagation delay time (t_{PHL}), A, B, C to W output, C_L = 50 pF	XV	S	2.0	16.5	2.0	16.5	2.0	18.0	2.0	18.0	ns
R31-R33	Propagation delay time (t_{PHL}),1-2A 1-2B, 3C to Y output, C_L = 50 pF	XV	S	2.0	23.0	2.0	23.0	2.0	25.0	2.0	25.0	ns
R34-R36	Propagation delay time (t_{PHL}), STROBE to W output, C_L = 50 PF	XV	S	2.0	14.0	2.0	14.0	2.0	15.0	2.0	15.0	ns
R37-R39	Propagation delay time (t_{PHL}), STROBE to Y output, C_L = 50 pF	XV	S	2.0	20.5	2.0	20.5	2.0	22.0	2.0	22.0	ns

See notes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE II. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
R40-R54	Propagation delay time (t_{PHL}), D0, D4-D7 to W output, C_L = 150 pF	XV	Q	4.0	10.5	4.0	10,5	4.0	11.5	4.0	11.5	ns
R55-R60	Propagation delay time (t_{PHL}), D1, D5 to Y output, C_L = 150 pF	XV	Q	4.0	15.5	4.0	15.5	4.0	17.0	4.0	17.0	ns
R61-R69	Propagation delay time (t_{PHL}), A, B, C to W output, C_L = 150 pF	XV	Q	4.0	18.0	4.0	18.0	4.0	20.0	4.0	20.0	ns
R70-R72	Propagation delay time (t_{PHL}),1-2A, 1-2B, 3C to Y output C_L = 150 pF	XV	Q	4.0	25.0	4.0	25.0	4.0	27.0	4.0	27.0	ns
R73-R75	Propagation delay time (t_{PHL}), STROBE to W output, C_L = 150 pF	XV	Q	4.0	16.0	4.0	16.0	4.0	17.0	4.0	17.0	ns
R76-R78	Propagation delay time (t_{PHL}), STROBE to Y output, C_L = 150 pF	XV	Q	4.0	22.5	4.0	22.5	4.0	24.0	4.0	24.0	ns
S1-S6	Output crosstalk, low level	XVI	Q	0	500	0	500	0	500	0	500	mV
T1-T6	Output crosstalk, high level	XVI	Q	0	400	0	400	0	400	0	400	mV
U	Power supply transient amplitude	XVII	Q	0	400	0	400	0	400	0	400	mV
V	Power supply filter capacitance	XVIII	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
W1-W28	Input capacitance, D0-D7, 3A, 3B, 3C, STROBE 3	XIX	Q	5.0	15.0	5.0	15.0	5.0	15.0	5.0	15.0	pF
W29-W32	Input capacitance, 1-2A, 1-2B, 1-2C, STROBE 1-2	XIX	Q	5.0	30.0	5.0	30.0	5.0	30.0	5.0	30.0	pF

NOTES:
1. (X) parameters shall be 100 percent tested during module acceptance tests.
2. (S) parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3. (Q) parameters shall be tested during qualification testing only.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:59Z
Check the source to verify that this is the current version before use.

4.3 Test equipment. The following test equipment is necessary to conduct acceptance tests for this module. Test equipment shall be as specified in MIL-STD-1665 and that listed below:

Equipment	MIL-STD-1665 designator	No. required (maximum per test figure)
DC Power supply	Item P01	4
Pulse generator	Item J01	1
Megohm bridge	Item R06	1
DC voltmeter	Item V03	1
DC ammeter	Item X01	1
Pulse generator	Item J05	1
Oscilloscope	Item D05	1
Oscilloscope probe	Item Y01	2
Capacitance meter	Item C05	1
Capacitance bridge	Item C01	1
Resistors		
249Ω ±1%, 1/4 W		6
51.1Ω ±1%, 1/4 W		1
1 kΩ ±5%, 1/4 W		1
51Ω ±5%, 1/2 W		2
590Ω ±1%, 1/4 W		1
Value as required for figure 14, note 2, 1/4 W		6
2.49 kΩ ±1%, 1/8 W		6
Capacitors		
Value as required for figure 14, note 4		6
Value as required for figure 14, note 5		6
Diode		
1N3600, 1N4150, or equivalent		6

All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table II

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program (SEMP) Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through W shall be performed in any order with the following exceptions. If conditions A, B, C, D, or E are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additons.

Source: https://assist.dla.mil -- Downloaded: 20260-20T15:59Z
Check the source to verify that this is the current version before use.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test condition A through W in table II shall be performed in the order specified in 4.4.2 using the initial limits at 25° ± 5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through V shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4.

4.5.2 Vibration tests. The vibrations tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 5. Following the vibration tests, the electrical tests of table II which are marked with an (X) in the test type column, and test condition V, shall be performed using the initial limits at 25° ± 5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through V shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table II Following the return of the module to 25°C, electrical tests D through W shall be performed in the order specified in 4 4.2 using the 25° ± 5°C end-of-life requirements of table II

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions Definitions are as specified in MIL-STD-1331

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table II, the module is considered to have completed its useful life.

6.3 Cross reference of classification Part number M28787/81-1 supersedes part number M28787/81. The two parts are interchangeable.

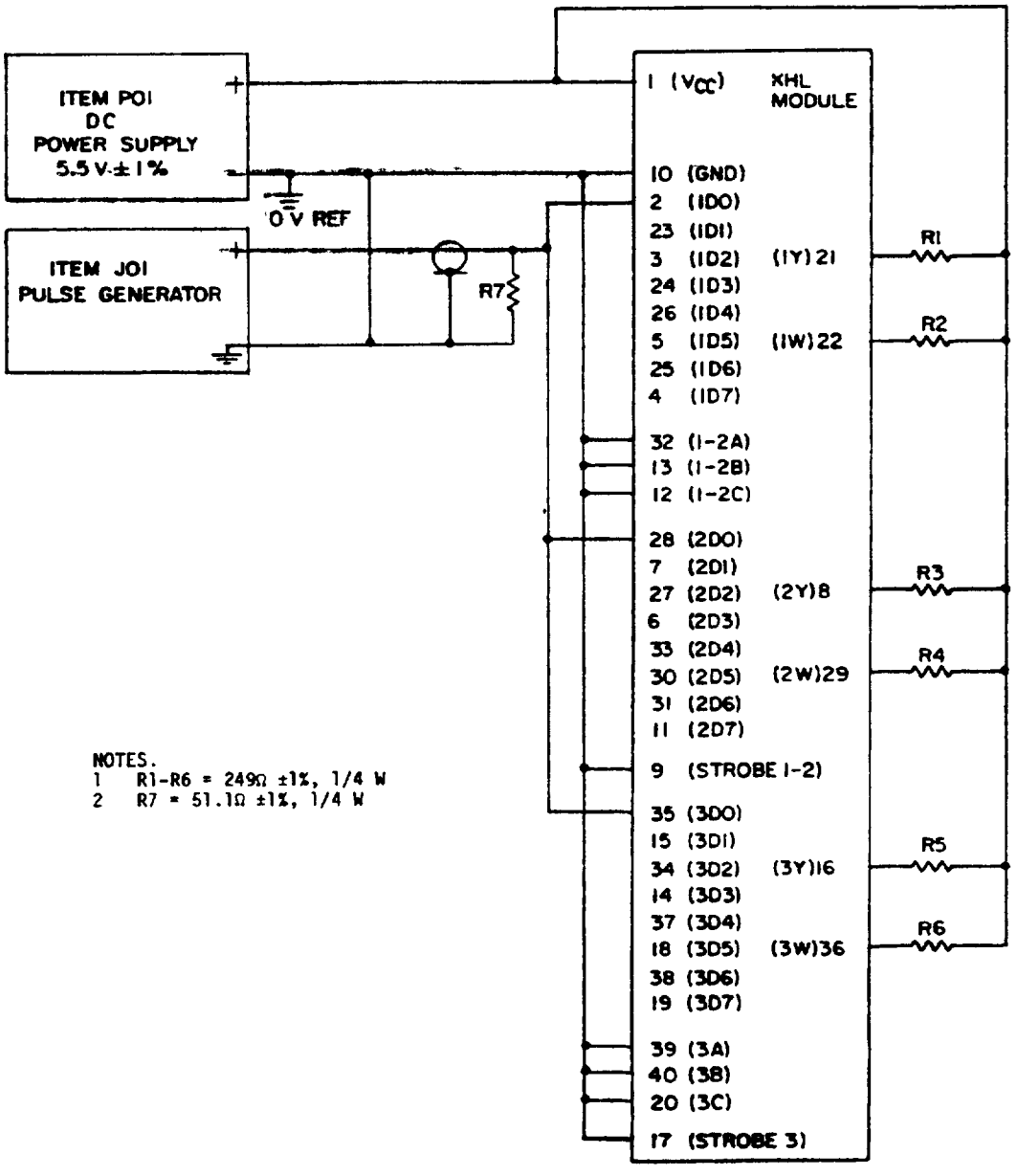
6 4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity
Navy - EC

Agent:
DLA - ES

(Project 5963-N031)

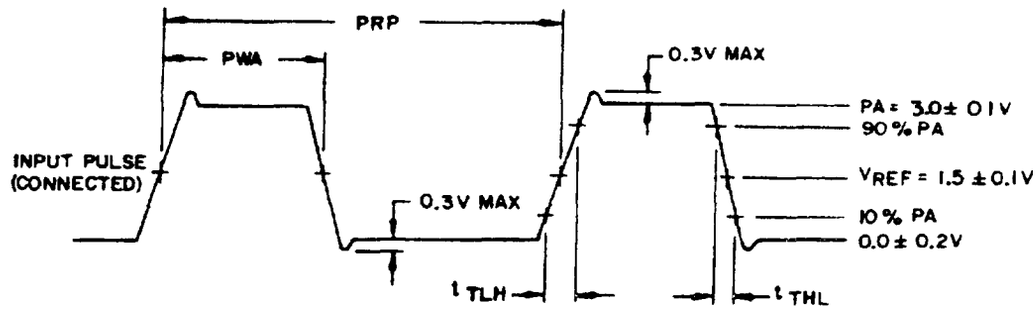
Source: https://assist.dla.mil -- Downloaded: 202640-20T15:59Z
Check the source to verify that this is the current version before use.



NOTES.
1 R1-R6 = 249Ω ±1%, 1/4 W
2 R7 = 51.1Ω ±1%, 1/4 W

FIGURE 2 Operating temperature and life test circuit

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:59Z
Check the source to verify that this is the current version before use.



NOTE Input pulse parameters
$t_{TLH} = t_{THL} \leq 10$ ns.
PWA = 500 ±100 ns
PRP = 1,000 ±200 ns

FIGURE 3. Operating temperature and life test circuit waveform

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:59Z
Check the source to verify that this is the current version before use.

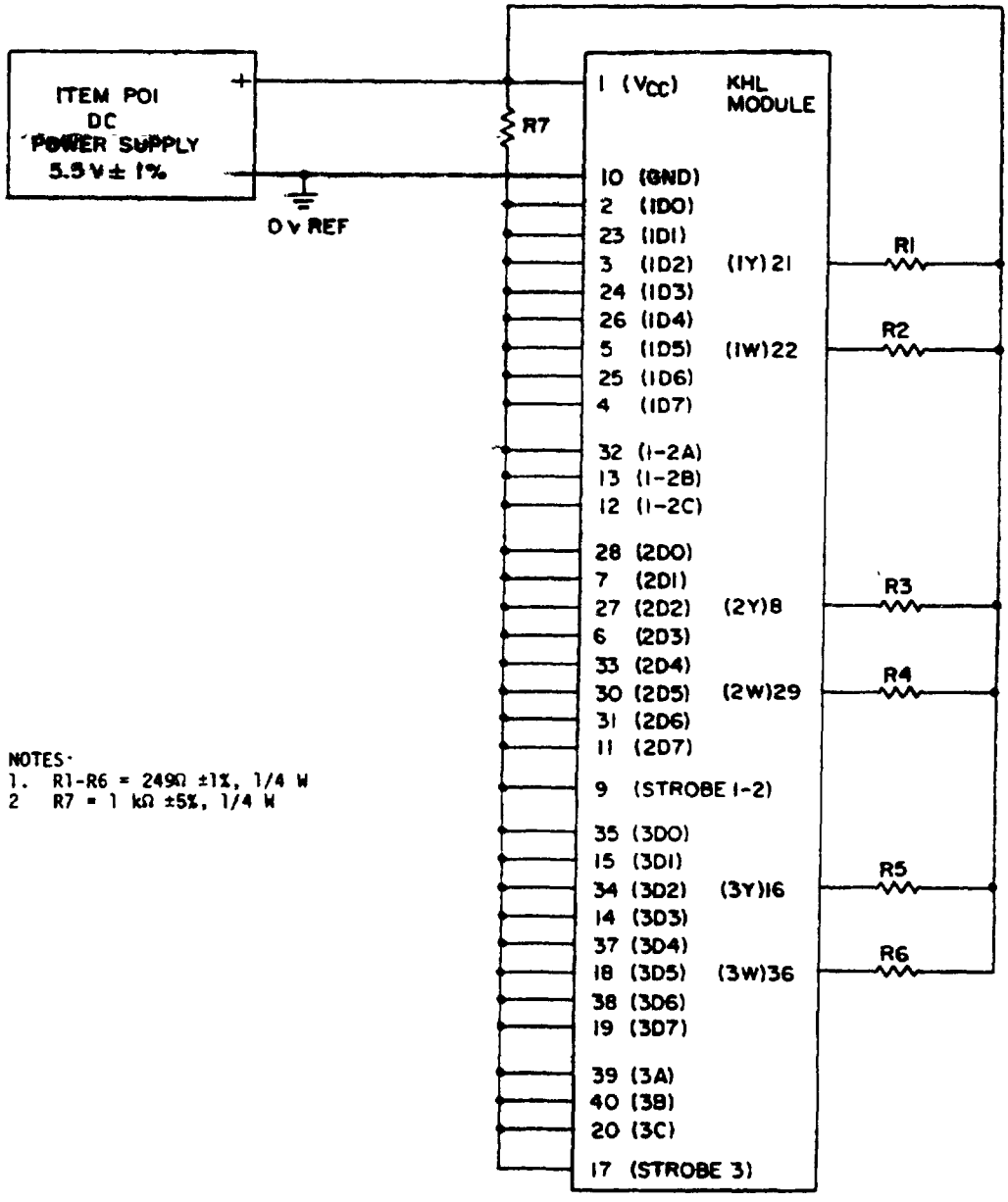


FIGURE 4 Component temperature test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:59Z
Check the source to verify that this is the current version before use.

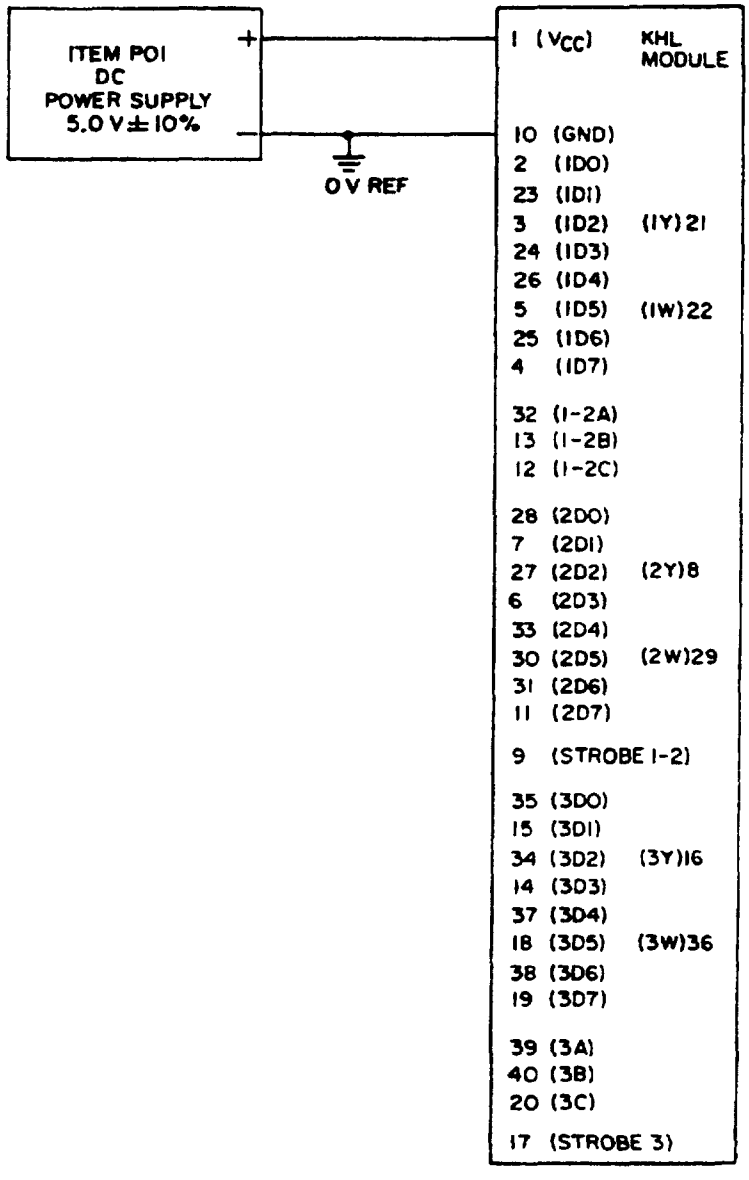


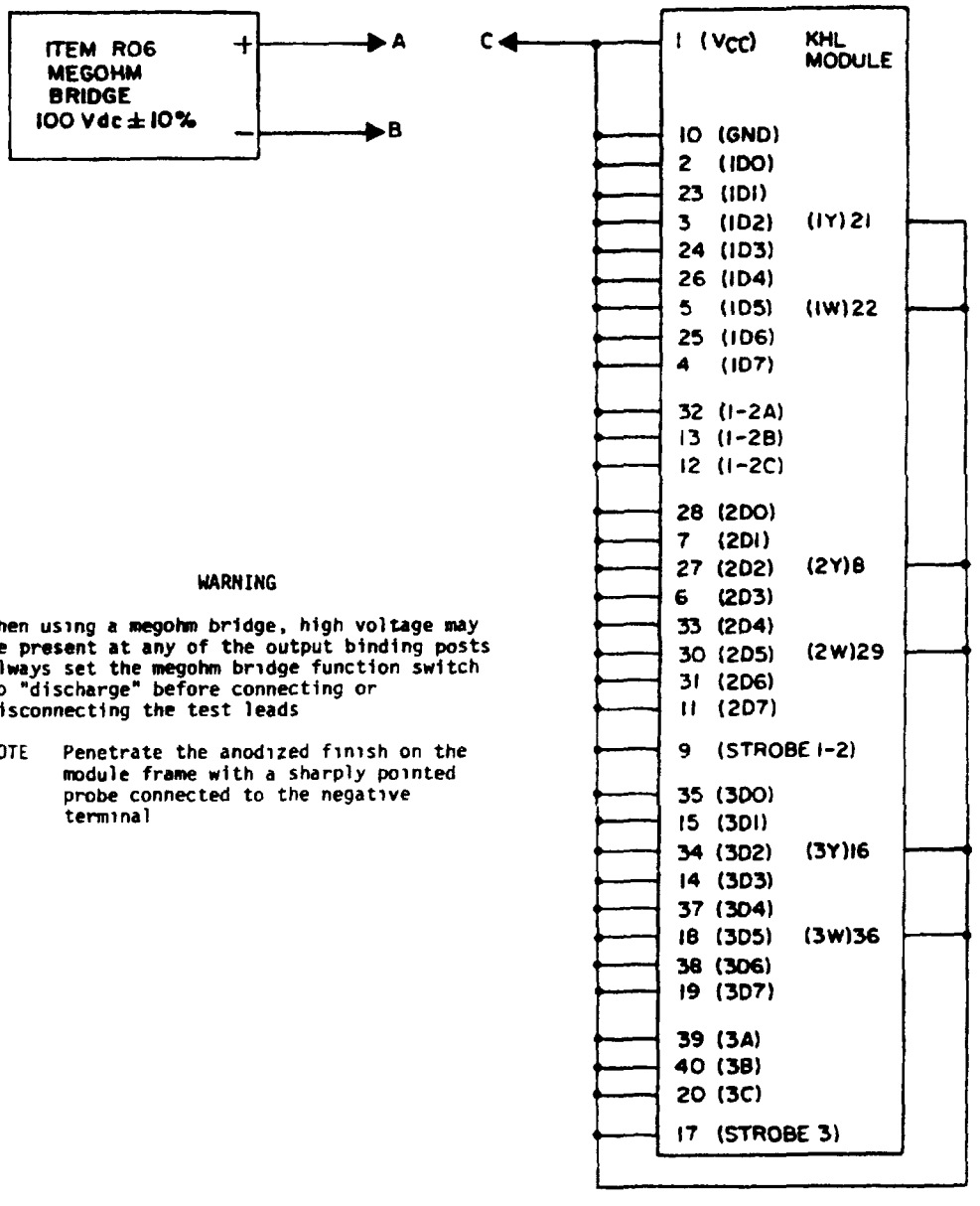
FIGURE 5 Vibration test circuit

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	C	Frame	10	-	MΩ	6

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:59Z
Check the source to verify that this is the current version before use.



WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads

NOTE Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal

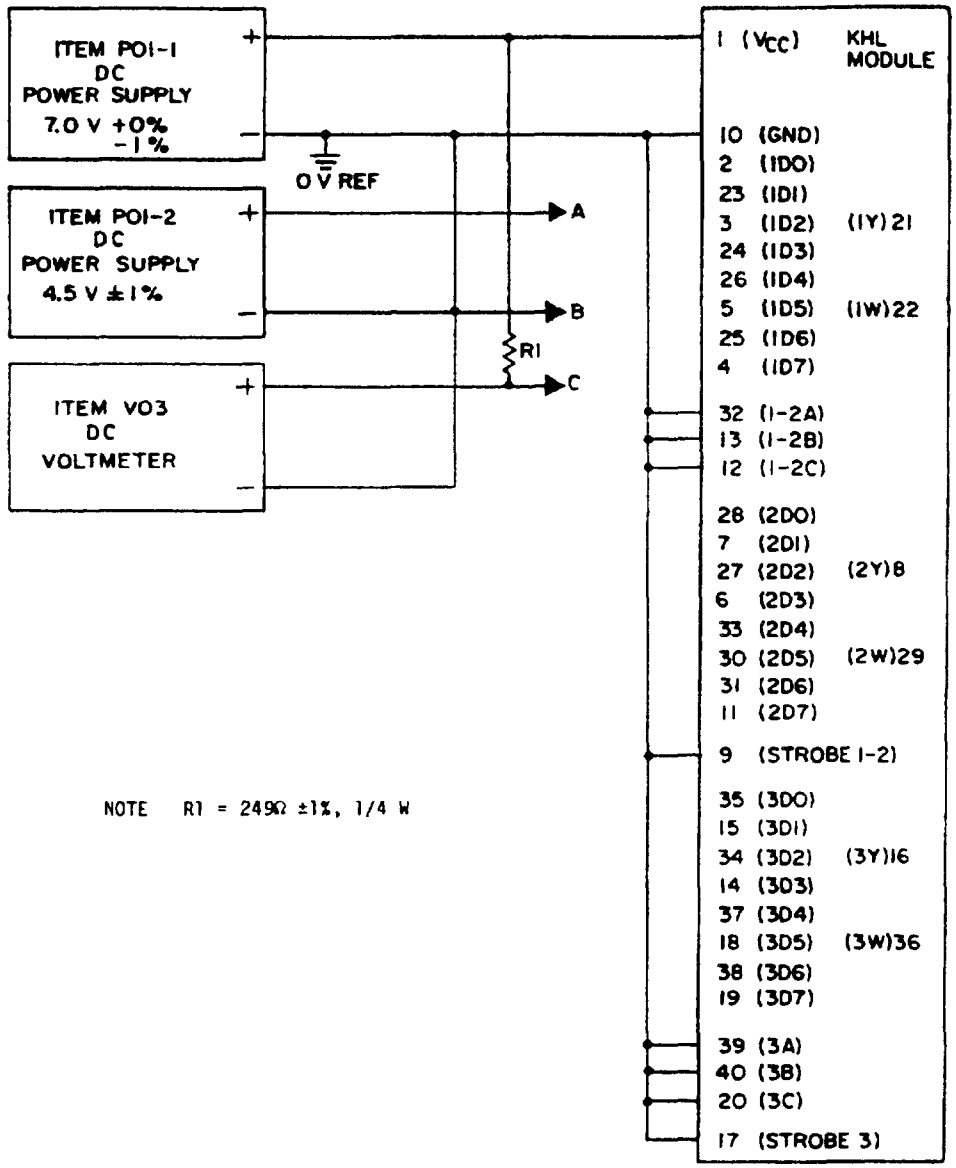
FIGURE 6 Isolation test circuit for table III

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE IV. V_{OUT} (low-loaded) test

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
B1	1Y V_{OUT} (low-loaded)	Open	2	21	0.01	1.00	V	7
B2	2Y	Open	28	8				
B3	3Y	Open	35	16				
B4	1W	2	Open	22				
B5	2W	28	Open	29				
B6	3W V_{OUT} (low-loaded)	35	Open	36	0.01	1.00	V	7

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T15:59Z
Check the source to verify that this is the current version before use.



NOTE R1 = 249Ω ±1%, 1/4 W

FIGURE 7 V_{OUT} (low-loaded) test circuit for table IV

Source: https://assist.dla.mil -- Downloaded: 20260-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE V. Input breakdown current test

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	1D0 Input breakdown current	3,4,5,6,7,11,12,13,14, 15,17,18,19,20,23,24,25, 26,27,28,30,31,32,33,34, 35,37,38,39,40	2	0.0	1.0	mA	8
C2	1D1	2,3,4,5,6,7,9,11,12,13, 14,15,17,18,19,20,24,25, 26,27,28,30,31,32,33,34, 35,37,38,39,40	23				
C3	1D2	2,4,5,6,7,9,11,12,13,14, 15,17,18,19,20,23,24,25, 26,27,28,30,31,32,33,34, 35,37,38,39,40	3				
C4	1D3	2,3,4,5,6,7,9,11,12,13, 14,15,17,18,19,20,23,25, 26,27,28,30,31,32,33,34, 35,37,38,39,40	24				
C5	1D4	2,3,4,5,6,7,9,11,12,13, 14,15,17,18,19,20,23,24, 25,27,28,30,31,32,33,34, 35,37,38,39,40	26				
C6	1D5	2,3,4,6,7,9,11,12,13,14, 15,17,18,19,20,23,24,25, 26,27,28,30,31,32,33,34, 35,37,38,39,40	5				
C7	1D6 Input breakdown current	2,3,4,5,6,7,9,11,12,13, 14,15,17,18,19,20,23,24, 26,27,28,30,31,32,33,34, 35,37,38,39,40	25	0.0	1.0	mA	8

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE V. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C8	1D7 Input breakdown current	2,3,5,6,7,9,11,12,13,14, 15,17,18,19,20,23,24,25, 26,27,28,30,31,32,33,34, 35,37,38,39,40	4	0.0	1.0	mA	8
C9	2D0	2,3,4,5,6,7,11,12,13,14, 15,17,18,19,20,23,24,25, 26,27,30,31,32,33,34,35, 37,38,39,40	28				
C10	2D1	2,3,4,5,6,9,11,12,13,14, 15,17,18,19,20,23,24,25, 26,27,28,30,31,32,33,34, 35,37,38,39,40	7				
C11	2D2	2,3,4,5,6,7,9,11,12,13, 14,15,17,18,19,20,23,24, 25,26,28,30,31,32,33,34, 35,37,38,39,40	27				
C12	2D3	2,3,4,5,7,9,11,12,13,14, 15,17,18,19,20,23,24,25, 26,27,28,30,31,32,33,34, 35,37,38,39,40	6				
C13	2D4	2,3,4,5,6,7,9,11,12,13, 14,15,17,18,19,20,23,24, 25,26,27,28,30,31,32,34, 35,37,38,39,40	33				
C14	2D5 Input breakdown current	2,3,4,5,6,7,9,11,12,13, 14,15,17,18,19,20,23,24, 25,26,27,28,31,32,33,34, 35,37,38,39,40	30	0.0	1.0	mA	8

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE V. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C15	2D6 Input breakdown current	2,3,4,5,6,7,9,11,12,13, 14,15,17,18,19,20,23,24, 25,26,27,28,30,32,33,34, 35,37,38,39,40	31	0.0	1.0	mA	8
C16	2D7	2,3,4,5,6,7,9,12,13,14, 15,17,18,19,20,23,24,25, 26,27,28,30,31,32,33,34, 35,37,38,39,40	11				
C17	3D0	2,3,4,5,6,7,9,11,12,13, 14,15,18,19,20,23,24,25, 26,27,28,30,31,32,33,34, 37,38,39,40	35				
C18	3D1	2,3,4,5,6,7,9,11,12,13, 14,17,18,19,20,23,24,25, 26,27,28,30,31,32,33,34, 35,37,38,39,40	15				
C19	3D2	2,3,4,5,6,7,9,11,12,13, 14,15,17,18,19,20,23,24, 25,26,27,28,30,31,32,33, 35,37,38,39,40	34				
C20	3D3	2,3,4,5,6,7,9,11,12,13, 15,17,18,19,20,23,24,25, 26,27,28,30,31,32,33,34, 35,37,38,39,40	14				
C21	3D4 Input breakdown current	2,3,4,5,6,7,9,11,12,13, 14,15,17,18,19,20,23,24, 25,26,27,28,30,31,32,33, 34,35,38,39,40	37	0.0	1.0	mA	8

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE V. Input breakdown current test - Continued.

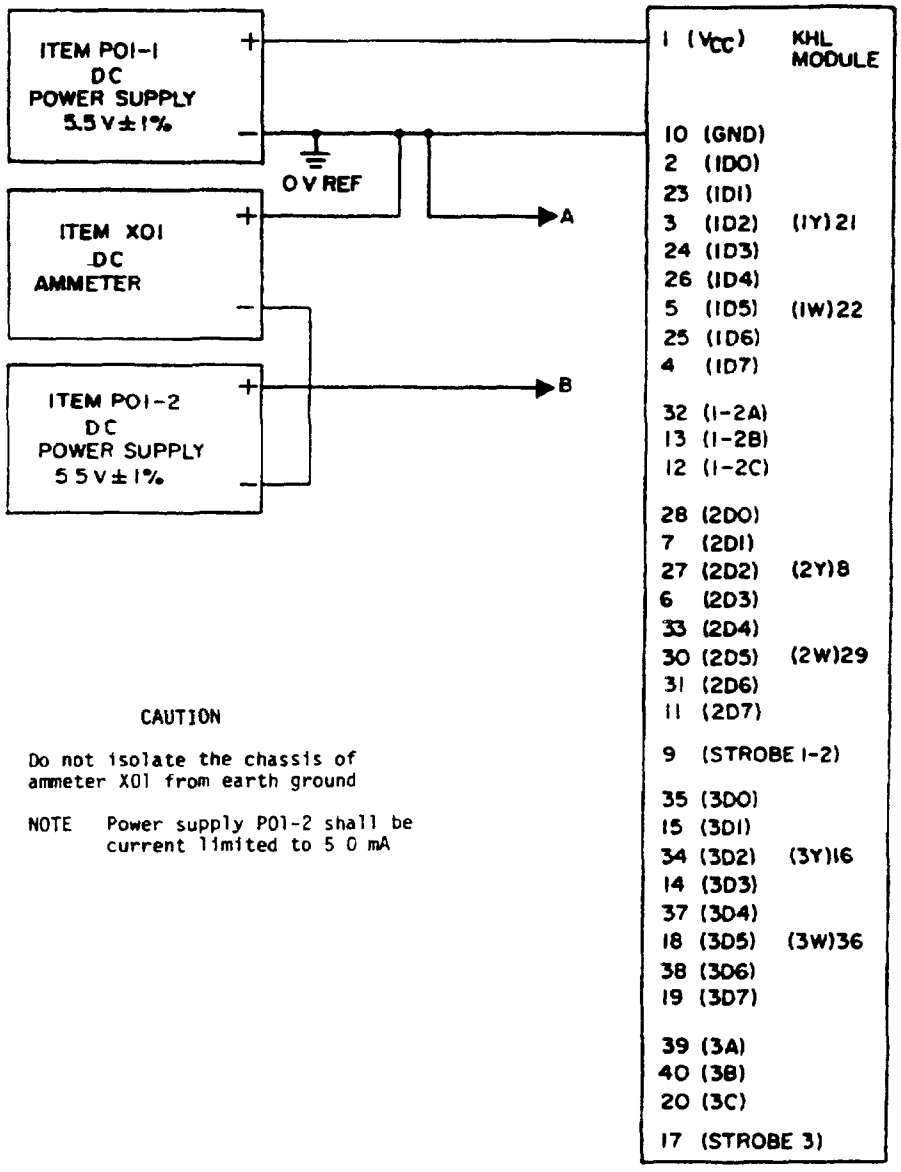
Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C22	3D5 Input breakdown current	2,3,4,5,6,7,9,11,12,13, 14,15,17,19,20,23,24,25, 26,27,28,30,31,32,33,34, 35,37,38,39,40	18	0.0	1.0	mA	8
C23	3D6	2,3,4,5,6,7,9,11,12,13, 14,15,17,18,19,20,23,24, 25,26,27,28,30,31,32,33, 34,35,37,39,40	38				
C24	3D7	2,3,4,5,6,7,9,11,12,13, 14,15,17,18,20,23,24,25, 26,27,28,30,31,32,33,34, 35,37,38,39,40	19				
C25	3A	2,3,4,5,6,7,9,11,12,13, 14,15,17,18,19,20,23,24, 25,26,27,28,30,31,32,33, 34,35,37,38,40	39				
C26	3B	2,3,4,5,6,7,9,11,12,13, 14,15,17,18,19,20,23,24, 25,26,27,28,30,31,32,33, 34,35,37,38,39	40				
C27	3C	2,3,4,5,6,7,9,11,12,13, 14,15,17,18,19,23,24,25, 26,27,28,30,31,32,33,34, 35,37,38,39,40	20				
C28	STROBE 3 Input breakdown current	2,3,4,5,6,7,9,11,12,13, 14,15,18,19,20,23,24,25, 26,27,28,30,31,32,33,34, 35,37,38,39,40	17	0.0	1.0	mA	8

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE V. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C29	1-2A Input breakdown current	2,3,4,5,6,7,9,11,12,13, 14,15,17,18,19,20,23,24, 25,26,27,28,30,31,33,34, 35,37,38,39,40	32	0.0	2.0	mA	8
C30	1-2B	2,3,4,5,6,7,9,11,12,14, 15,17,18,19,20,23,24,25, 26,27,28,30,31,32,33,34, 35,37,38,39,40	13				
C31	1-2C	2,3,4,5,6,7,9,11,13,14, 15,17,18,19,20,23,24,25, 26,27,28,30,31,32,33,34, 35,37,38,39,40	12				
C32	STROBE 1-2 Input breakdown current	2,3,4,5,6,7,11,12,13,14, 15,17,18,19,20,23,24,25, 26,27,28,30,31,32,33,34, 35,37,38,39,40	9	0.0	2.0	mA	8

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T15:59Z
Check the source to verify that this is the current version before use.



CAUTION

Do not isolate the chassis of ammeter XO1 from earth ground

NOTE Power supply PO1-2 shall be current limited to 5.0 mA

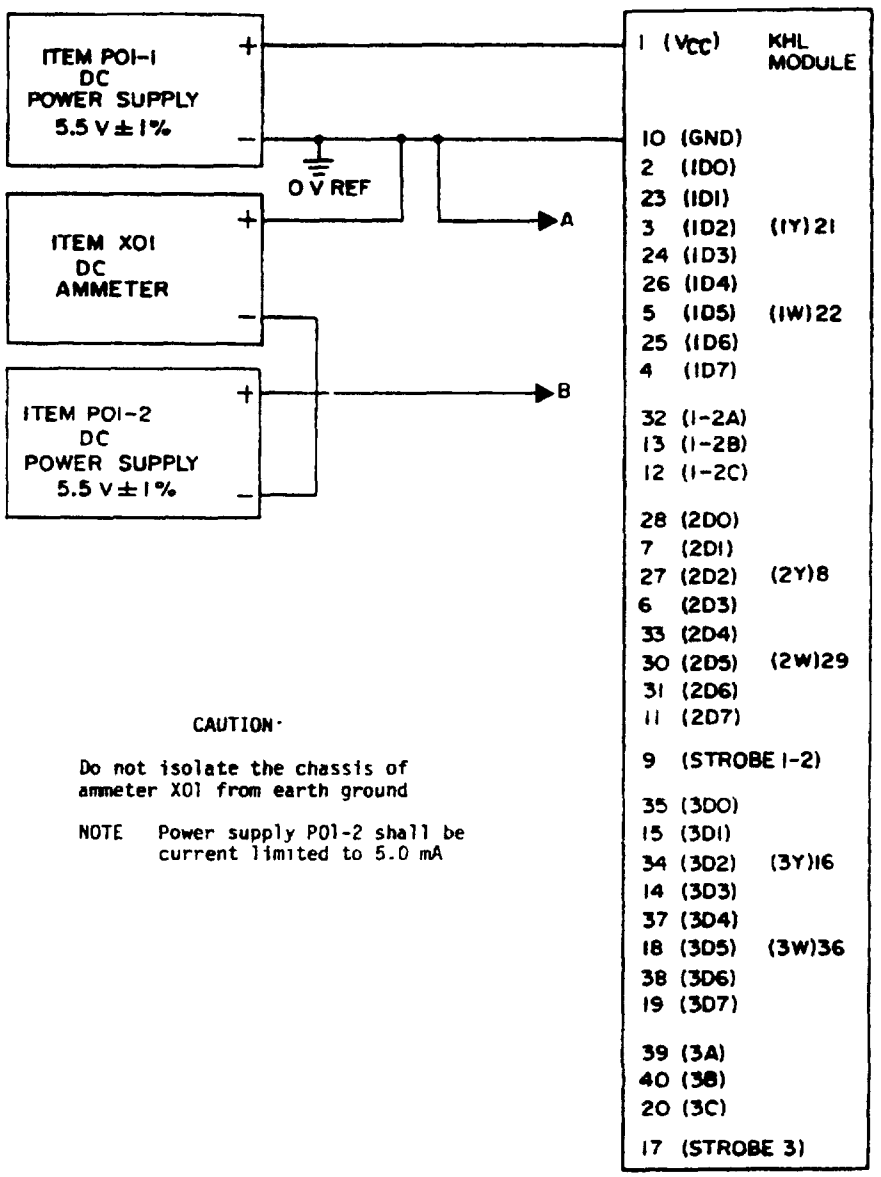
FIGURE 8 Input breakdown current test circuit for table V

Source: https://assist.dla.mil -- Downloaded: 20260-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE VI. Output leakage current test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	1Y Output leakage current	9	21	0.0	1.0	mA	9
D2	2Y	9	8				
D3	3Y	17	16				
D4	1W	Open	22				
D5	2W	Open	29				
D6	3W Output leakage current	Open	36	0.0	1.0	mA	9

Source: https://assist.dla.mil -- Downloaded: 20260-20T15:59Z
Check the source to verify that this is the current version before use.



CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground

NOTE Power supply PO1-2 shall be current limited to 5.0 mA

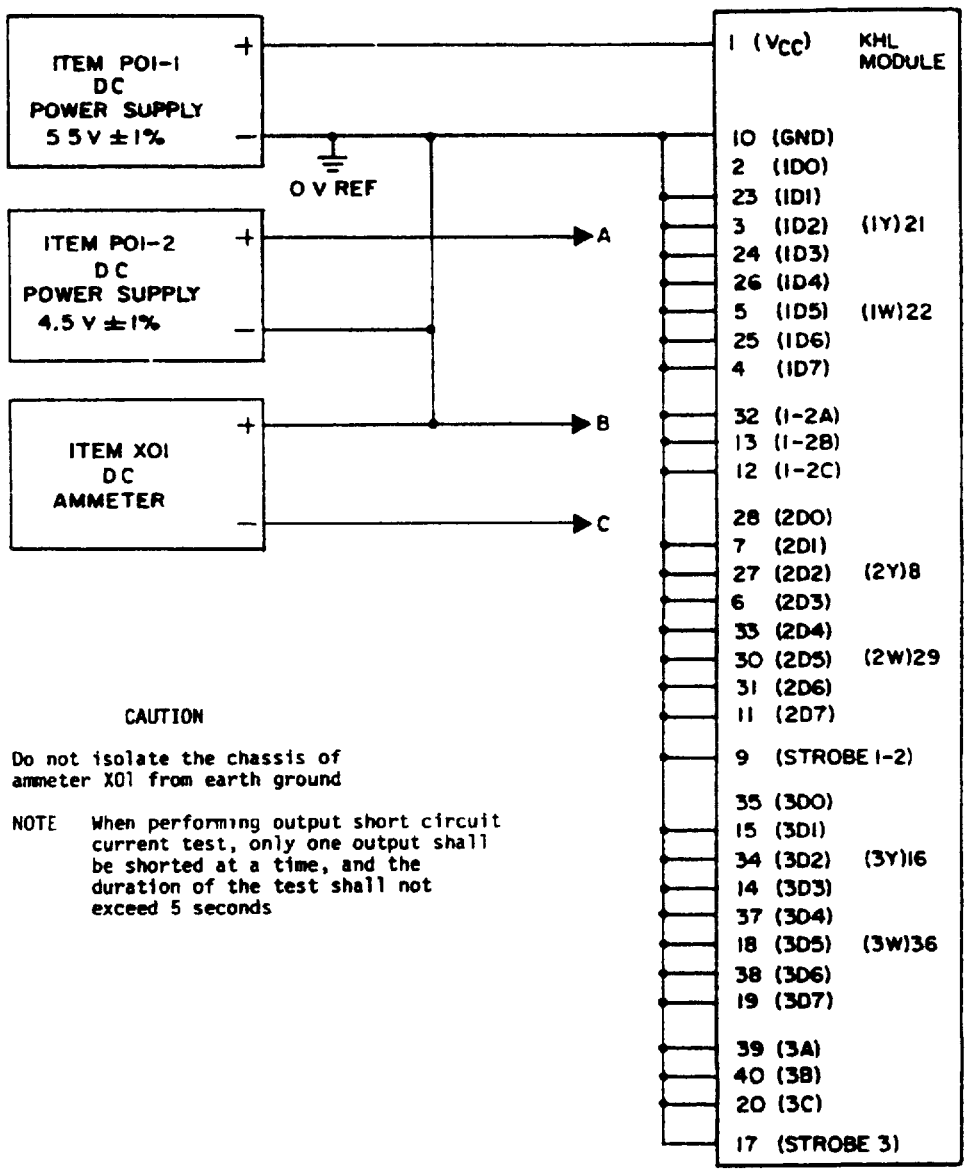
FIGURE 9 Output leakage current test circuit for table VI

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE VII Output short circuit current test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
E1	1Y Output short circuit current	2	28,35	21	-40	-100	mA	10
E2	2Y	28	2,35	8				
E3	3Y	35	2,28	16				
E4	1W	Open	2,28,35	22				
E5	2W	Open	2,28,35	29				
E6	3W Output short circuit current	Open	2,28,35	36	-40	-100	mA	10

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:59Z
Check the source to verify that this is the current version before use.



CAUTION

Do not isolate the chassis of ammeter X01 from earth ground

NOTE When performing output short circuit current test, only one output shall be shorted at a time, and the duration of the test shall not exceed 5 seconds

FIGURE 10 Output short circuit current test for table VII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE VIII. Functional test.

Test cond	Parameter	Connect			Output logic states						Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	1Y	2Y	3Y	1W	2W	3W	Min	Max		
F1	Functional, STROBE high	2,3,4,5, 6,7,9,11, 14,15,17, 18,19,23, 24,25,26, 27,28,30, 31,33,34, 35,37,38	12,13, 20,32, 39,40	See notes 2 and 3	L	L	L	H	H	H	See note 1			11
F2	Functional, STROBE high	2,3,4,5, 6,7,9,11, 14,15,17, 18,19,23, 24,25,26, 27,28,30, 31,32,33, 34,35,37, 38,39	12,13 20,40	See notes 2 and 3	L	L	L	H	H	H	See note 1			11
F3	Functional, STROBE high	2,3,4,5, 6,7,9,11, 13,14,15, 17,18,19, 23,24,25, 26,27,28, 30,31,33, 34,35,37, 38,40	12,20, 32,39	See notes 2 and 3	L	L	L	H	H	H	See note 1			11

See notes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE VIII. Functional tests - Continued

Test cond	Parameter	Connect			Output logic states						Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	1Y	2Y	3Y	1W	2W	3W	Min	Max		
F4	Functional, STROBE high	2,3,4,5, 6,7,9,11, 13,14,15, 17,18,19, 23,24,25, 26,27,28, 30,31,32, 33,34,35, 37,38,39, 40	12,20	See notes 2 and 3	L	L	L	H	H	H	See note 1			11
F5	Functional, STROBE high	2,3,4,5, 6,7,9,11, 12,14,15, 17,18,19, 20,23,24, 25,26,27, 28,30,31, 33,34,35, 37,38	13,32, 39,40	See notes 2 and 3	L	L	L	H	H	H	See note 1			11
F6	Functional, STROBE high	2,3,4,5, 6,7,9,11, 12,14,15, 17,18,19, 20,23,24, 25,26,27, 28,30,31, 32,33,34, 35,37,38, 39	13,40	See notes 2 and 3	L	L	L	H	H	H	See note 1			11

See notes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE VIII. Functional test - Continued

Test cond	Parameter	Connect			Output logic states						Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	1Y	2Y	3Y	1W	2W	3W	Min	Max		
F7	Functional, STROBE high	2,3,4,5, 6,7,9,11, 12,13,14, 15,17,18, 19,20,23, 24,25,26, 27,28,30, 31,33,34, 35,37,38, 40	32,39	See notes 2 and 3	L	L	L	H	H	H	See note 1			11
F8	Functional, STROBE high	2,3,4,5, 6,7,9,11, 12,13,14, 15,17,18, 19,20,23, 24,25,26, 27,28,30, 31,32,33, 34,35,37, 38,39,40	Open	See notes 2 and 3	L	L	L	H	H	H	See note 1			11
F9	Functional, DO low	3,4,5,6, 7,11,14, 15,18,19, 23,24,25, 26,27,30, 31,33,34, 37,38	2,9,12, 13,17, 20,28, 32,35, 39,40	See notes 2 and 3	L	L	L	H	H	H	See note 1			11

See notes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE VIII. Functional test - Continued.

Test cond	Parameter	Connect			Output logic states						Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	1Y	2Y	3Y	1W	2W	3W	Min	Max		
F10	Functional, D0 high	2,28,35	3,4,5,6, 7,9,11, 12,13,14, 15,17,18, 19,20,23, 24,25,26, 27,30,31, 32,33,34, 37,38,39, 40	See notes 2 and 3	H	H	H	L	L	L	See note 1			11
F11	Functional, D1 low	2,3,4,5, 6,11,14, 18,19,24, 25,26,27, 28,30,31, 32,33,34, 35,37,38, 39	7,9,12, 13,15,17, 20,23,40	See notes 2 and 3	L	L	L	H	H	H	See note 1			11
F12	Functional, D1 high	7,15,23, 32,39	2,3,4,5, 6,9,11,12, 13,14,17, 18,19,20, 24,25,26, 27,28,30, 31,33,34, 35,37,38, 40	See notes 2 and 3	H	H	H	L	L	L	See note 1			11

See notes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE VIII. Functional test - Continued.

Test cond	Parameter	Connect			Output logic states						Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	1Y	2Y	3Y	1W	2W	3W	Min	Max		
F13	Functional, D2 low	2,4,5,6, 7,11,13, 14,15,18, 19,23,24, 25,26,28, 30,31,33, 35,37,38, 40	3,9,12, 17,20,27, 32,34,39	See notes 2 and 3	L	L	L	H	H	H	See note 1			11
F14	Functional, D2 high	3,13,27, 34,40	2,4,5,6, 7,9,11, 12,14,15, 17,18,19, 20,23,24, 25,26,28, 30,31,32, 33,35,37, 38,39	See notes 2 and 3	H	H	H	L	L	L	See note 1			11
F15	Functional, D3 low	2,3,4,5, 7,11,13, 15,18,19, 23,25,26, 27,28,30, 31,32,33, 34,35,37, 38,39,40	6,9,12, 14,17,20, 24	See notes 2 and 3	L	L	L	H	H	H	See note 1			11

See notes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE VIII. Functional test - Continued.

Test cond	Parameter	Connect			Output logic states						Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	1Y	2Y	3Y	1W	2W	3W	Min	Max		
F16	Functional, D3 high	6,13,14, 24,32,39, 40	2,3,4,5, 7,9,11, 12,15,17, 18,19,20, 23,25,26, 27,28,30, 31,33,34, 35,37,38	See notes 2 and 3	H	H	H	L	L	L	See note 1			11
F17	Functional, D4 low	2,3,4,5, 6,7,11, 12,14,15, 18,19,20, 23,24,25, 27,28,30, 31,34,35, 38	9,13,17, 26,32,33, 37,39,40	See notes 2 and 3	L	L	L	H	H	H	See note 1			11
F18	Functional, D4 high	12,20,26, 33,37	2,3,4,5, 6,7,9,11, 13,14,15, 17,18,19, 23,24,25, 27,28,30, 31,32,34, 35,38,39, 40	See notes 2 and 3	H	H	H	L	L	L	See note 1			11

See notes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE VIII. Functional test - Continued.

Test cond	Parameter	Connect			Output logic states						Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	1Y	2Y	3Y	1W	2W	3W	Min	Max		
F19	Functional, D5 low	2,3,4,6,7, 11,12,14, 15,19,20, 23,24,25, 26,27,28, 31,32,33, 34,35,37, 38,39	5,9,13, 17,18, 30,40	See notes 2 and 3	L	L	L	H	H	H	See note 1			11
F20	Functional D5 high	5,12,18, 20,30,32, 39	2,3,4,6, 7,9,11, 13,14,15, 17,19,23, 24,25,26, 27,28,31, 33,34,35, 37,38,40	See notes 2 and 3	H	H	H	L	L	L	See note 1			11
F21	Functional, D6 low	2,3,4,5, 6,7,11, 12,13,14, 15,18,19, 20,23,24, 26,27,28, 30,33,34, 35,37,40	9,17,25, 31,32,38, 39	See notes 2 and 3	L	L	L	H	H	H	See note 1			11

See notes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE VIII. Functional test - Continued.

Test cond	Parameter	Connect			Output logic states						Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	1Y	2Y	3Y	1W	2W	3W	Min	Max		
F22	Functional, D6 high	12,13,20, 25,31,38, 40	2,3,4,5, 6,7,9,11, 14,15,17, 18,19,23, 24,26,27, 28,30,32, 33,34,35, 37,39	See notes 2 and 3	H	H	H	L	L	L	See note 1			11
F23	Functional, D7 low	2,3,5,6, 7,12,13, 14,15,18, 20,23,24, 25,26,27, 28,30,31, 32,33,34, 35,37,38, 39,40	4,9,11, 17,19	See notes 2 and 3	L	L	L	H	H	H	See note 1			11
F24	Functional, D7 high	4,11,12, 13,19,20, 32,39,40	2,3,5,6, 7,9,14, 15,17,18, 23,24,25, 26,27,28, 30,31,33, 34,35,37, 38	See notes 2 and 3	H	H	H	L	L	L	See note 1			11

See notes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE VIII. Functional test - Continued.

Test cond	Parameter	Connect			Output logic states						Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	1Y	2Y	3Y	1W	2W	3W	Min	Max		
F25	Functional, D0, D5, D6 high	2,5,12, 18,20,25, 28,30,31, 35,38	3,4,6,7, 9,11,13, 14,15,17, 19,23,24, 26,27,32, 33,34,37, 39,40	See notes 2 and 3	L	L	L	H	H	H	See note 1			11

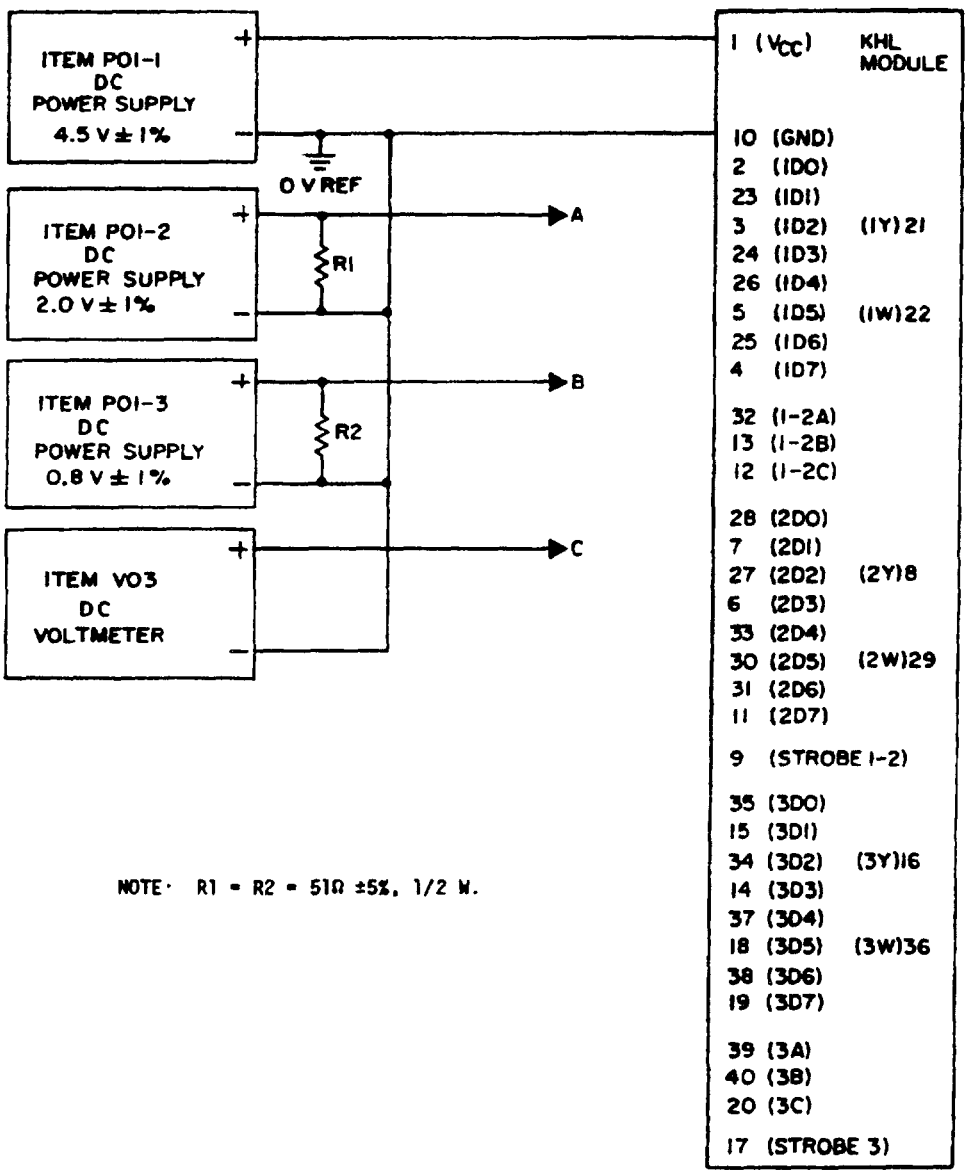
NOTES:

1. Output logic states:
   L = Low-voltage level = 0.0 to 0.5 V.
   H = High-voltage level = 2.5 to 4.5 V.

2. For test conditions F1-F25, verification of the output logic states in note 1 is obtained by connecting voltmeter (V03), point C, to each output pin in succession after input conditions have been accomplished.

3. 1Y, 2Y, 3Y, 1W, 2W, 3W output logic states are measured at output pins 21, 8, 16, 22,29, and 36 respectively.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:59Z
Check the source to verify that this is the current version before use.



NOTE: R1 = R2 = 51Ω ±5%, 1/2 W.

FIGURE 11. Functional test circuit for table VIII.

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE IX. Input clamp diode voltage test.

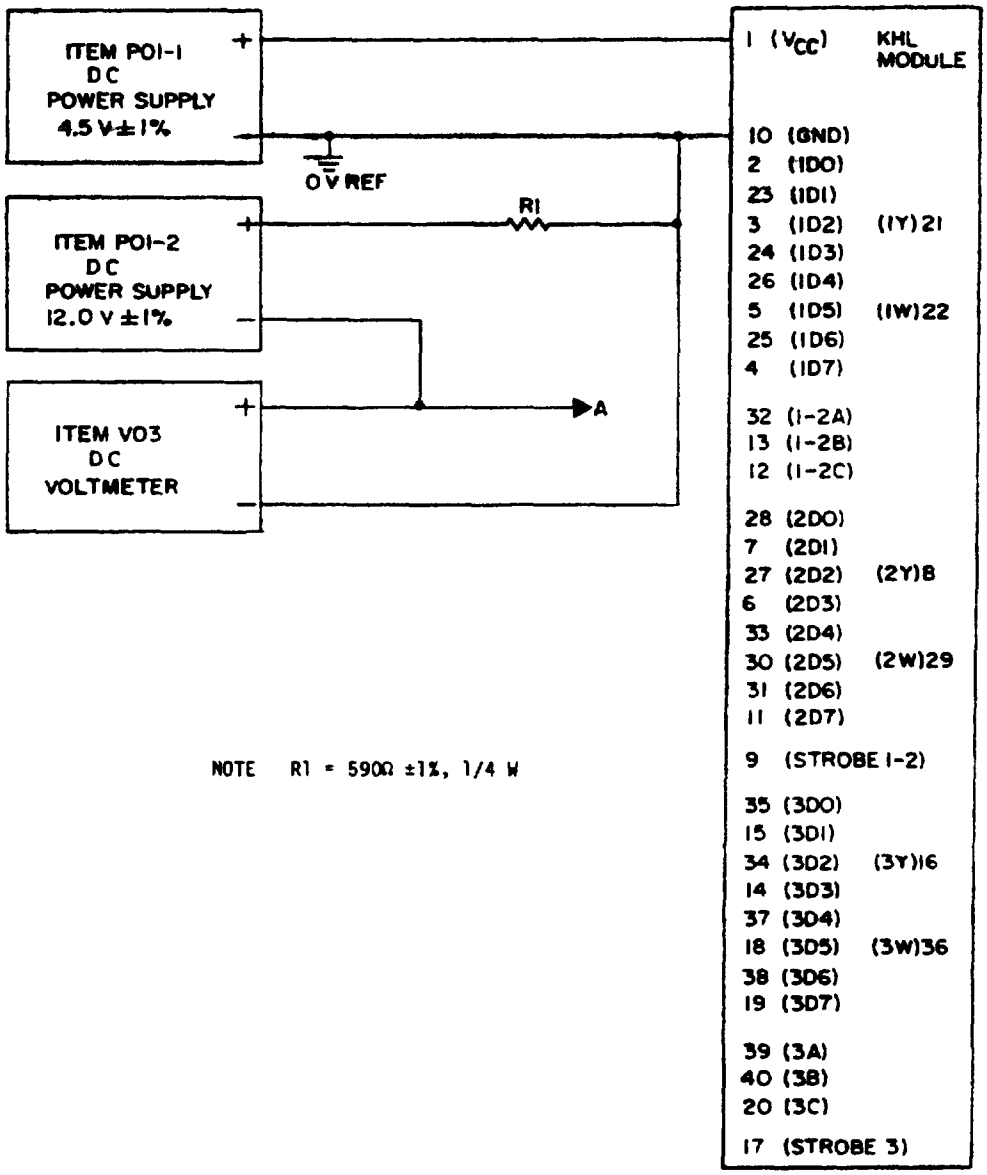
Test condition	Parameter		Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	Min	Max		
G1	1D0	Input clamp diode voltage	2	-0.4	-1.2	V	12
G2	1D1		23				
G3	1D2		3				
G4	1D3		24				
G5	1D4		26				
G6	1D5		5				
G7	1D6		25				
G8	1D7		4				
G9	2D0		28				
G10	2D1		7				
G11	2D2		27				
G12	2D3		6				
G13	2D4		33				
G14	2D5		30				
G15	2D6		31				
G16	2D7		11				
G17	3D0		35				
G18	3D1	Input clamp diode voltage	15	-0.4	-1.2	V	12

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE IX. Input clamp diode voltage test - Continued.

Test condition	Parameter		Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	Min	Max		
G19	3D2	Input clamp diode voltage	34	-0.4	-1.2	V	12
G20	3D3		14				
G21	3D4		37				
G22	3D5		18				
G23	3D6		38				
G24	3D7		19				
G25	1-2A		32				
G26	1-2B		13				
G27	1-2C		12				
G28	3A		39				
G29	3B		40				
G30	3C		20				
G31	STROBE 1-2		9				
G32	STROBE 3	Input clamp diode voltage	17	-0.4	-1.2	V	12

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T15:59Z
Check the source to verify that this is the current version before use.



NOTE R1 = 590Ω ±1%, 1/4 W

FIGURE 12 Input clamp diode voltage test circuit for table IX

Source: https://assist.dla.mil -- Downloaded: 20261O-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE X. Static output voltage (low-level) tests

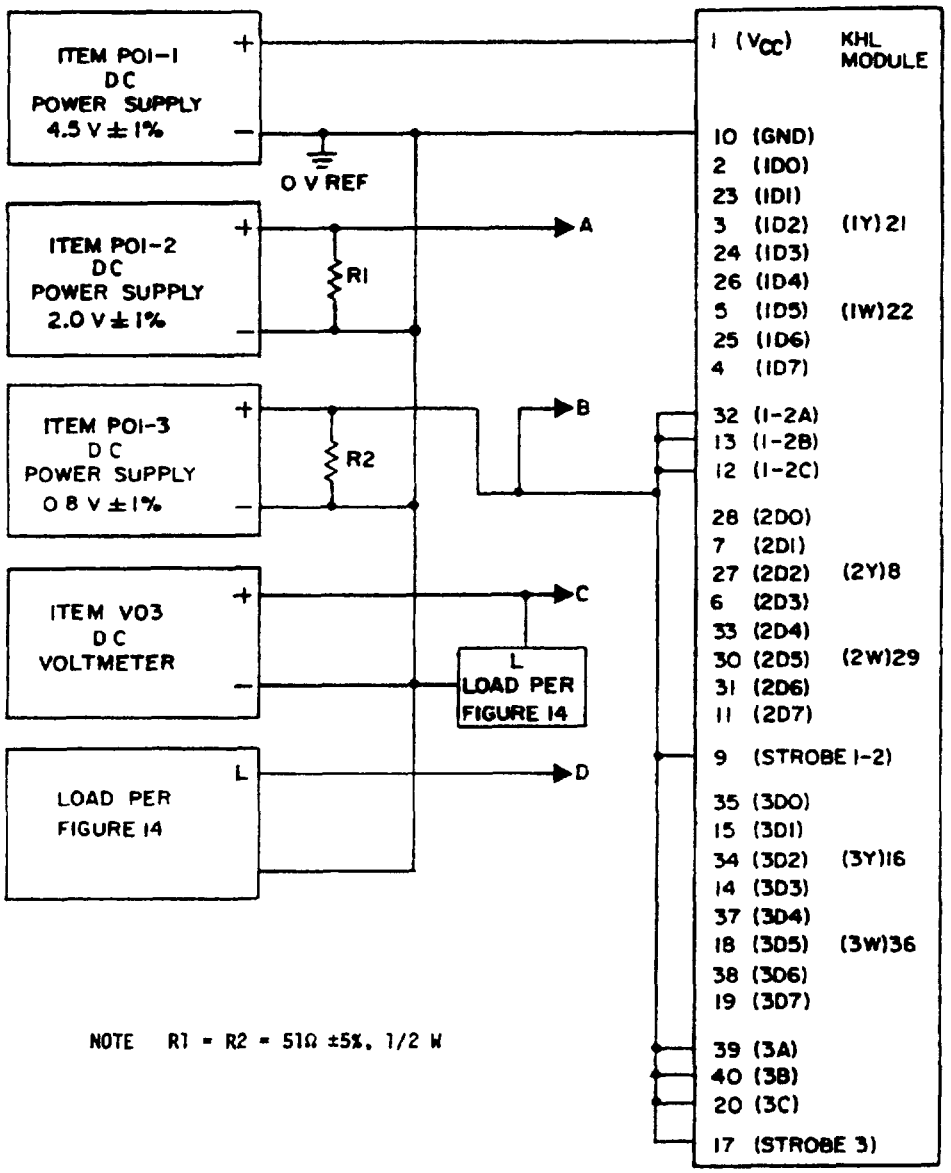
Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25° ±5°C Ambient Min	Initial limits 25° ±5°C Ambient Max	Unit	Test figure
H1	1Y Static output voltage, low level	Open	2,28, 35	21	22	0.0	0.5	V	13
H2	2Y	Open	2,28, 35	8	29				
H3	3Y	Open	2,28, 35	16	36				
H4	1W	2,28, 35	Open	22	21				
H5	2W	2,28, 35	Open	29	8				
H6	3W Static output voltage, low level	2,28, 35	Open	36	16	0.0	0.5	V	13

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE XI. Static output voltage (high-level) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
J1	1Y Static output voltage, high level	2	28,35	21	22	2.5	4.5	V	13
J2	2Y	28	2,35	8	29				
J3	3Y	35	2,28	16	36				
J4	1W	28,35	2	22	21				
J5	2W	2,35	28	29	8				
J6	3W Static output voltage, high level	2,28	35	36	16	2 5	4.5	V	13

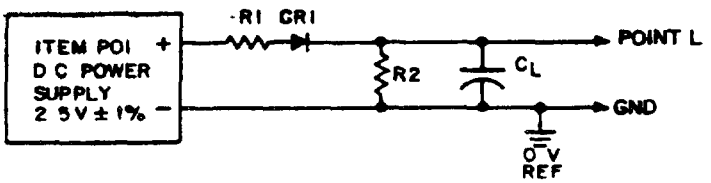
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:59Z
Check the source to verify that this is the current version before use.



NOTE R1 = R2 = 51Ω ±5%, 1/2 W

FIGURE 13 Static output voltage (low-level and high-level) test circuit for tables X and XI

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:59Z
Check the source to verify that this is the current version before use.



NOTES:
1. CR1 = 1N3600, 1N4150, or equivalent.
2. R1 = Select value for a current flow of 20 mA ±5%, out of point L with point L held at 0.5 V.
3. R2 = 2.49 kΩ ±1%, 1/8 W.
4. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions N1-N6, P1-P6, Q1-Q39, R1-R39, S1-S6, and T1-T6.
5 C_L = 150 ±5 pF (including probe and parasitic capacitance) for test conditions N7-N12, P7-P12, Q40-Q78, and R40-R78.
6 Capacitor C_L is not required for dc electrical test conditions H1-H6 and J1-J6
7 Lead length from load circuit to module output pin under test is to be less than 1 inch (25.4 mm).

FIGURE 14. KHL logic load circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE XII Input current (high-level) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	1D0 Input current, high level	3,4,5,6,7,11,12,13,14, 15,17,18,19,20,23,24,25, 26,27,28,30,31,32,33,34, 35,37,38,39,40	2	0	50	µA	15
K2	1D1	2,3,4,5,6,7,9,11,12,13, 14,15,17,18,19,20,24,25, 26,27,28,30,31,32,33,34, 35,37,38,39,40	23				
K3	1D2	2,4,5,6,7,9,11,12,13,14, 15,17,18,19,20,23,24,25, 26,27,28,30,31,32,33,34, 35,37,38,39,40	3				
K4	1D3	2,3,4,5,6,7,9,11,12,13, 14,15,17,18,19,20,23,25, 26,27,28,30,31,32,33,34, 35,37,38,39,40	24				
K5	1D4	2,3,4,5,6,7,9,11,12,13, 14,15,17,18,19,20,23,24, 25,27,28,30,31,32,33,34, 35,37,38,39,40	26				
K6	1D5	2,3,4,6,7,9,11,12,13,14, 15,17,18,19,20,23,24,25, 26,27,28,30,31,32,33,34, 35,37,38,39,40	5				
K7	1D6 Input current high level	2,3,4,5,6,7,9,11,12,13, 14,15,17,18,19,20,23,24, 26,27,28,30,31,32,33,34, 35,37,38,39,40	25	0	50	µA	15

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE XII. Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K8	1D7 Input current, high level	2,3,5,6,7,9,11,12,13,14, 15,17,18,19,20,23,24,25, 26,27,28,30,31,32,33,34, 35,37,38,39,40	4	0	50	µA	15
K9	2D0	2,3,4,5,6,7,11,12,13,14, 15,17,18,19,20,23,24,25, 26,27,30,31,32,33,34,35, 37,38,39,40	28				
K10	2D1	2,3,4,5,6,9,11,12,13,14, 15,17,18,19,20,23,24,25, 26,27,28,30,31,32,33,34, 35,37,38,39,40	7				
K11	2D2	2,3,4,5,6,7,9,11,12,13, 14,15,17,18,19,20,23,24, 25,26,28,30,31,32,33,34, 35,37,38,39,40	27				
K12	2D3	2,3,4,5,7,9,11,12,13,14, 15,17,18,19,20,23,24,25, 26,27,28,30,31,32,33,34, 35,37,38,39,40	6				
K13	2D4	2,3,4,5,6,7,9,11,12,13, 14,15,17,18,19,20,23,24, 25,26,27,28,30,31,32,34, 35,37,38,39,40	33				
K14	2D5 Input current, high level	2,3,4,5,6,7,9,11,12,13, 14,15,17,18,19,20,23,24, 25,26,27,28,31,32,33,34, 35,37,38,39,40	30	0	50	µA	15

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE XII. Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K15	2D6 Input current, high level	2,3,4,5,6,7,9,11,12,13, 14,15,17,18,19,20,23,24, 25,26,27,28,30,32,33,34, 35,37,38,39,40	31	0	50	µA	15
K16	2D7	2,3,4,5,6,7,9,12,13,14, 15,17,18,19,20,23,24,25, 26,27,28,30,31,32,33,34, 35,37,38,39,40	11				
K17	3D0	2,3,4,5,6,7,9,11,12,13, 14,15,18,19,20,23,24,25, 26,27,28,30,31,32,33,34, 37,38,39,40	35				
K18	3D1	2,3,4,5,6,7,9,11,12,13, 14,17,18,19,20,23,24,25, 26,27,28,30,31,32,33,34, 35,37,38,39,40	15				
K19	3D2	2,3,4,5,6,7,9,11,12,13, 14,15,17,18,19,20,23,24, 25,26,27,28,30,31,32,33, 35,37,38,39,40	34				
K20	3D3	2,3,4,5,6,7,9,11,12,13, 15,17,18,19,20,23,24,25, 26,27,28,30,31,32,33,34, 35,37,38,39,40	14				
K21	3D4 Input current, high level	2,3,4,5,6,7,9,11,12,13, 14,15,17,18,19,20,23,24, 25,26,27,28,30,31,32,33, 34,35,38,39,40	37	0	50	µA	15

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE XII. Input current (high-level) test - Continued.

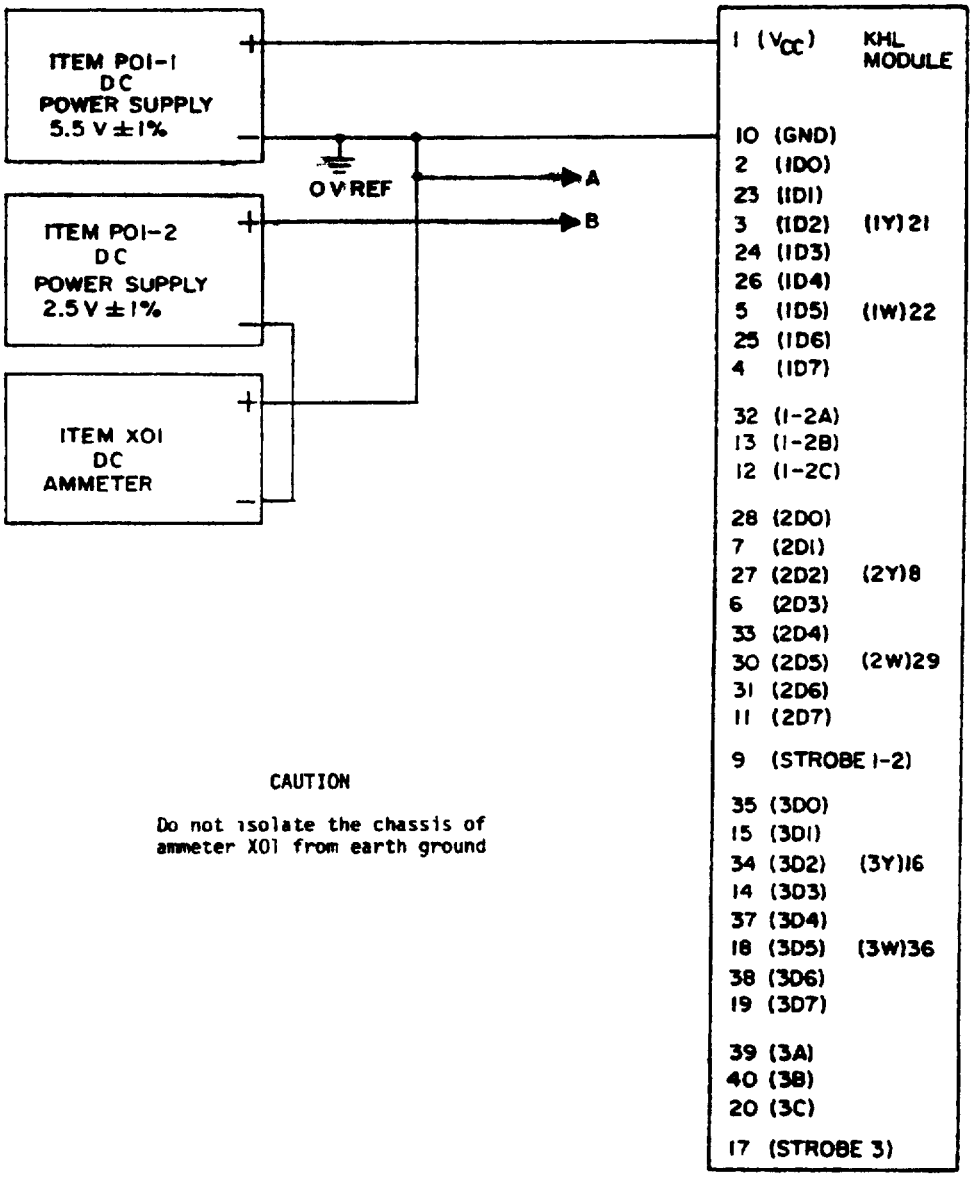
Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K22	3D5 Input current, high level	2,3,4,5,6,7,9,11,12,13, 14,15,17,19,20,23,24,25, 26,27,28,30,31,32,33,34, 35,37,38,39,40	18	0	50	µA	15
K23	3D6	2,3,4,5,6,7,9,11,12,13, 14,15,17,18,19,20,23,24, 25,26,27,28,30,31,32,33, 34,35,37,39,40	38				
K24	3D7	2,3,4,5,6,7,9,11,12,13, 14,15,17,18,20,23,24,25, 26,27,28,30,31,32,33,34, 35,37,38,39,40	19				
K25	3A	2,3,4,5,6,7,9,11,12,13, 14,15,17,18,19,20,23,24, 25,26,27,28,30,31,32,33, 34,35,37,38,40	39				
K26	3B	2,3,4,5,6,7,9,11,12,13, 14,15,17,18,19,20,23,24, 25,26,27,28,30,31,32,33, 34,35,37,38,39	40				
K27	3C	2,3,4,5,6,7,9,11,12,13, 14,15,17,18,19,23,24,25, 26,27,28,30,31,32,33,34, 35,37,38,39,40	20				
K28	STROBE 3 Input current, high level	2,3,4,5,6,7,9,11,12,13, 14,15,18,19,20,23,24,25, 26,27,28,30,31,32,33,34, 35,37,38,39,40	17	0	50	µA	15

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE XII. Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K29	1-2A Input current, high level	2,3,4,5,6,7,9,11,12,13, 14,15,17,18,19,20,23,24, 25,26,27,28,30,31,33,34, 35,37,38,39,40	32	0	100	µA	15
K30	1-2B	2,3,4,5,6,7,9,11,12,14, 15,17,18,19,20,23,24,25, 26,27,28,30,31,32,33,34, 35,37,38,39,40	13				
K31	1-2C	2,3,4,5,6,7,9,11,13,14, 15,17,18,19,20,23,24,25, 26,27,28,30,31,32,33,34, 35,37,38,39,40	12				
K32	STROBE 1-2 Input current, high level	2,3,4,5,6,7,11,12,13,14, 15,17,18,19,20,23,24,25, 26,27,28,30,31,32,33,34, 35,37,38,39,40	9	0	100	µA	15

Source: https://assist.dla.mil -- Downloaded: 20260-20T15:59Z
Check the source to verify that this is the current version before use.



CAUTION

Do not isolate the chassis of ammeter XOI from earth ground

FIGURE 15 Input current (high-level) test circuit for table XII.

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE XIII. Input current (low-level) test.

Test condition	Parameter		Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	Min	Max		
L1	1D0	Input current, low level	2	-0.1	-2.2	mA	16
L2	1D1		23				
L3	1D2		3				
L4	1D3		24				
L5	1D4		26				
L6	1D5		5				
L7	1D6		25				
L8	1D7		4				
L9	2D0		28				
L10	2D1		7				
L11	2D2		27				
L12	2D3		6				
L13	2D4		33				
L14	2D5		30				
L15	2D6		31				
L16	2D7		11				
L17	3D0		35				
L18	3D1	Input current, low level	15	-0.1	-2.2	mA	16

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE XIII. Input current (low-level) test - Continued)

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
L19	3D2 Input current, low level	34	-0.1	-2.2	mA	16
L20	3D3	14				
L21	3D4	37				
L22	3D5	18				
L23	3D6	38				
L24	3D7	19				
L25	3A	39				
L26	3B	40				
L27	3C	20				
L28	STROBE 3	17	-0.1	-2.2		
L29	1-2A	32	-0.2	-4.4		
L30	1-2B	13				
L31	1-2C	12				
L32	STROBE 1-2 Input current, low level	9	-0.2	-4.4	mA	16

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:59Z
Check the source to verify that this is the current version before use.

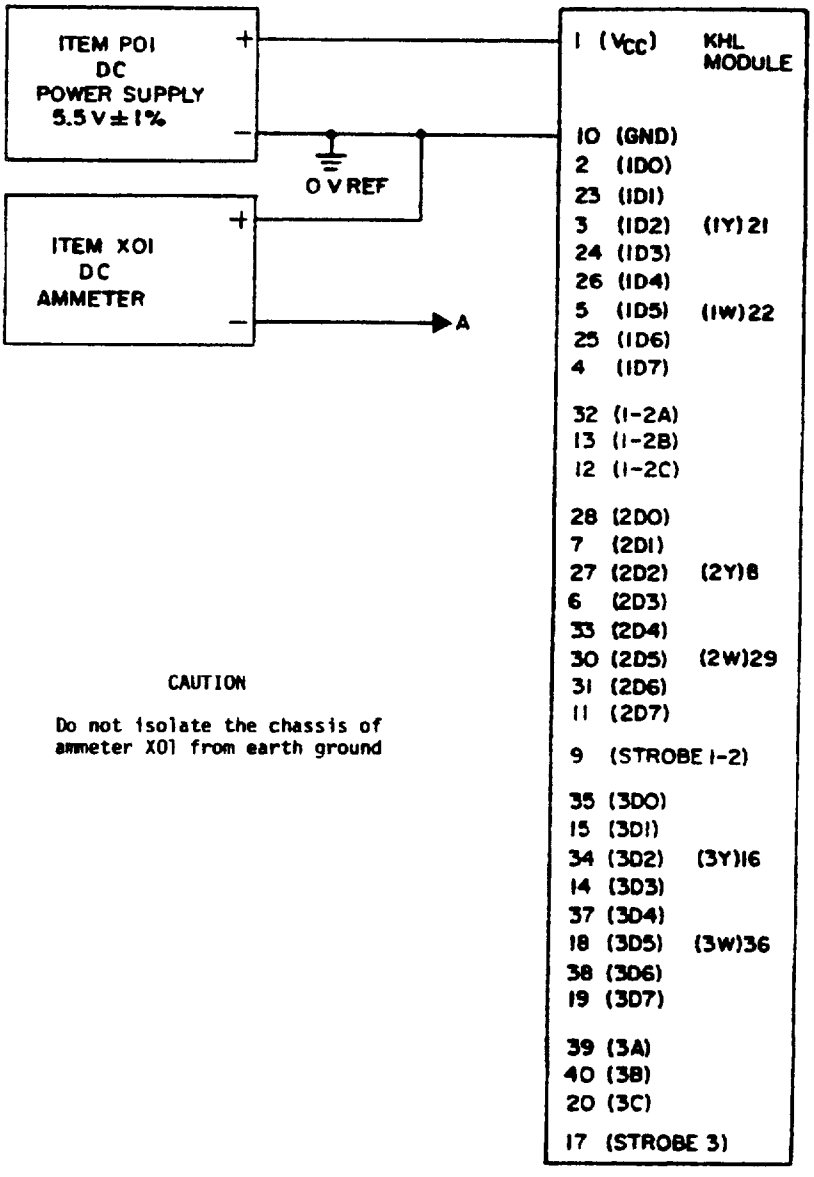


FIGURE 16. Input current (low-level) test circuit for table XIII

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE XIV. Power supply current test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
M	Power supply current	45	210	mA	17

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:59Z
Check the source to verify that this is the current version before use.

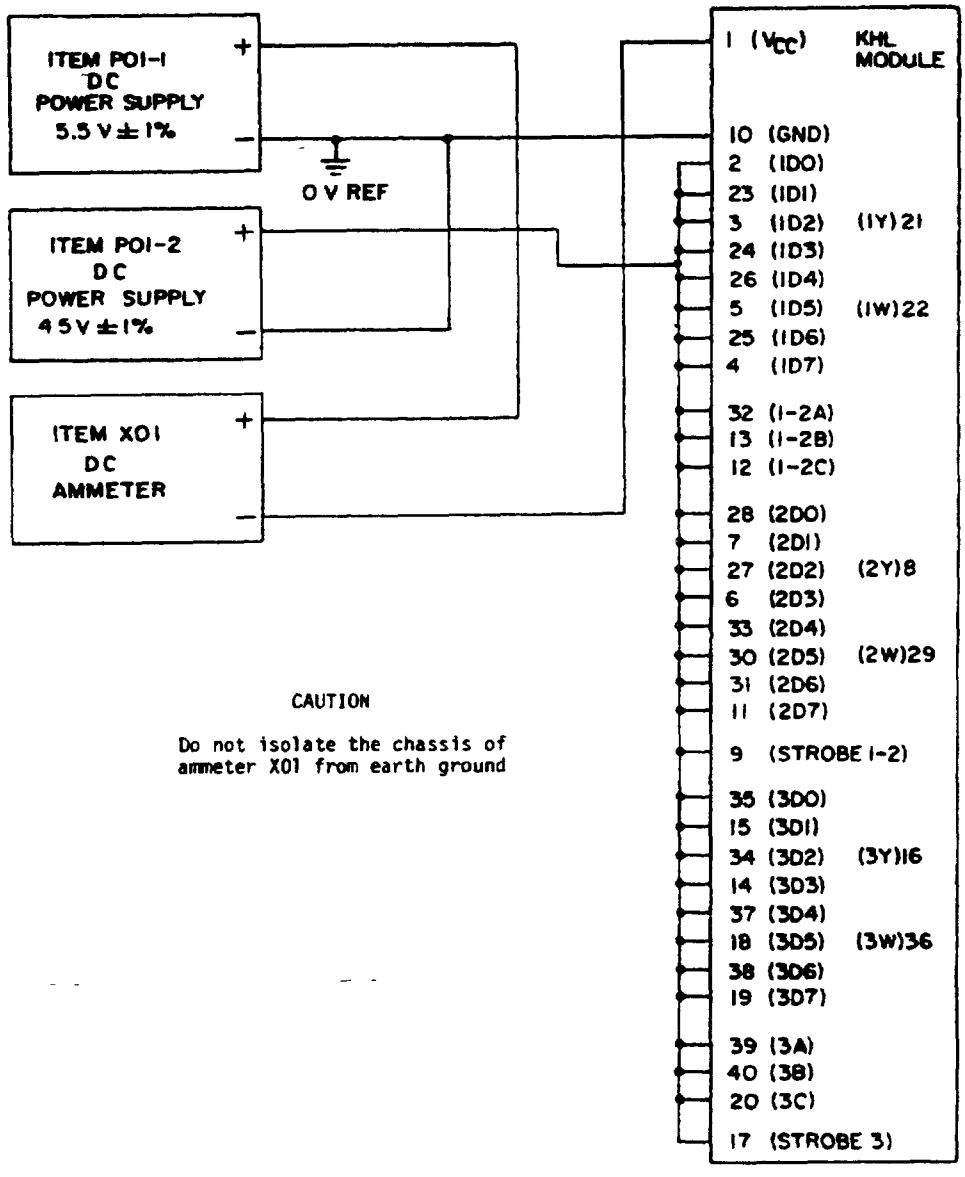


FIGURE 17 Power supply current test circuit for table XIV

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test.-

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
N1	1Y	Transition time (t_{TLH}), C_L = 50 pF	Open	9,12, 13,32	2	21	2.0	15.0	ns	18,19
N2	1W				2	22		12.0		
N3	2Y				28	8		15.0		
N4	2W			9,12, 13,32	28	29		12.0		
N5	3Y			17,20, 39,40	35	16		15.0		
N6	3W	Transition time (t_{TLH}), C_L = 50 pF		17,20, 39,40	35	36		12.0		
N7	1Y	Transition time (t_{TLH}), C_L = 150 pF		9,12, 13,32	2	21		27.0		
N8	1W				2	22		24.0		
N9	2Y				28	8		27.0		
N10	2W			9,12, 13,32	28	29		24.0		
N11	3Y			17,20, 39,40	35	16		27.0		
N12	3W	Transition time (t_{TLH}), C_L = 150 pF	Open	17,20, 39,40	35	36	2.0	24.0	ns	18,19

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P1	1Y	Transition time (t_{THL}), C_L = 50 pF	Open	9,12, 13,32	2	21	2.0	6.0	ns	18,19
P2	1W				2	22				
P3	2Y				28	8				
P4	2W			9,12, 13,32	28	29				
P5	3Y			17,20, 39,40	35	16				
P6	3W	Transition time (t_{THL}), C_L = 50 pF		17,20, 39,40	35	36		6.0		
P7	1Y	Transition time (t_{THL}), C_L = 150 pF		9,12, 13,32	2	21		11.0		
P8	1W				2	22				
P9	2Y				28	8				
P10	2W			9,12, 13,32	28	29				
P11	3Y			17,20, 39,40	35	16				
P12	3W	Transition time (t_{THL}), C_L = 150 pF	Open	17,20, 39,40	35	36	2.0	11.0	ns	18,19

Source: https://assist.dla.mil -- Downloaded: 202640-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q1	1D0 to 1W Propagation delay time (t_{PLH}), C_L = 50 pF	Open	9,12, 13,32	2	22	2.0	8.5	ns	18,22
Q2	1D1 to 1W	32	9,12, 13	23					
Q3	1D2 to 1W	13	9,12, 32	3					
Q4	1D3 to 1W	13,32	9,12	24					
Q5	1D4 to 1W	12	9,13, 32	26	22				
Q6	2D0 to 2W	Open	9,12, 13,32	28	29				
Q7	2D1 to 2W	32	9,12, 13	7					
Q8	2D2 to 2W	13	9,12, 32	27					
Q9	2D3 to 2W	13,32	9,12	6					
Q10	2D4 to 2W	12	9,13, 32	33	29				
Q11	3D0 to 3W	Open	17,20, 39,40	35	36				
Q12	3D1 to 3W	39	17,20, 40	15	36				
Q13	3D2 to 3W Propagation delay time (t_{PLH}), C_L = 50 pF	40	17,20, 39	34	36	2.0	8.5	ns	18,22

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q14	3D3 to 3W Propagation delay time (t_{PLH}), C_L = 50 pF	39,40	17,20	14	36	2.0	8.5	ns	18,22
Q15	3D4 to 3W	20	17,39, 40	37	36		8.5		18,22
Q16	1D1 to 1Y	32	9,12, 13	23	21		13.5		18,21
Q17	1D5 to 1Y	12,32	9,13	5	21				
Q18	2D1 to 2Y	32	9,12, 13	7	8				
Q19	2D5 to 2Y	12,32	9,13	30	8				
Q20	3D1 to 3Y	39	17,20, 40	15	16				
Q21	3D5 to 3Y	20,39	17,40	18	16		13.5		18,21
Q22	1-2A to 1W	2	9,12, 13,23	32	22		16.5		18,20
Q23	1-2B to 1W	5,12, 32	4,9	13	22				
Q24	1-2C to 1W	13,24, 32	4,9	12	22				
Q25	1-2A to 2W	28	7,9, 12,13	32	29				
Q26	1-2B to 2W Propagation delay time (t_{PLH}), C_L = 50 pF	12,30, 32	9,11	13	29	2.0	16.5	ns	18,20

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q27	1-2C to 2W Propagation delay time (t_{PLH}), C_L = 50 pF	6,13, 32	9,11	12	29	2.0	16.5	ns	18,20
Q28	3A to 3W	35	15,17, 20,40	39	36				
Q29	3B to 3W	18,20, 39	17,19	40	36				
Q30	3C to 3W	14,39, 40	17,19	20	36		16.5		18,20
Q31	1-2A to 1Y	23	2,9, 12,13	32	21		23.0		18,21
Q32	1-2B to 2Y	11,12, 32	9,30	13	8		23.0		18,21
Q33	3C to 3Y	19,39, 40	14,17	20	16		23.0		18,21
Q34	STROBE 1-2 to 1W	2	12,13, 32	9	22		14.0		18,20
Q35	STROBE 1-2 to 2W	28	12,13, 32	9	29		14.0		18,20
Q36	STROBE 3 to 3W	35	20,39, 40	17	36		14.0		18,20
Q37	STROBE 1-2 to 1Y	2	12,13, 32	9	21		20.5		18,22
Q38	STROBE 1-2 to 2Y Propagation delay time (t_{PLH}), C_L = 50 pF	28	12,13, 32	9	8	2.0	20.5	ns	18,22

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q39	STROBE 3 to 3Y	Propagation delay time (t_{PLH}), C_L = 50 pF	35	20,39, 40	17	16	2.0	20.5	ns	18,22
Q40	1D0 to 1W	Propagation delay time (t_{PLH}), C_L = 150 pF	Open	9,12, 13,32	2	22	4.0	10.5		
Q41	1D1 to 1W		32	9,12, 13	23					
Q42	1D2 to 1W		13	9,12, 32	3					
Q43	1D3 to 1W		13,32	9,12	24					
Q44	1D4 to 1W		12	9,13, 32	26	22				
Q45	2D0 to 2W		Open	9,12, 13,32	28	29				
Q46	2D1 to 2W		32	9,12, 13	7					
Q47	2D2 to 2W		13	9,12, 32	27					
Q48	2D3 to 2W		13,32	9,12	6					
Q49	2D4 to 2W		12	9,13, 32	33	29				
Q50	3D0 to 3W		Open	17,20, 39,40	35	36				
Q51	3D1 to 3W	Propagation delay time (t_{PLH}), C_L = 150 pF	39	17,20, 40	15	36	4.0	10.5	ns	18,22

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q52	3D2 to 3W Propagation delay time (t_{PLH}), C_L = 150 pF	40	17,20, 39	34	36	4.0	10.5	ns	18,22
Q53	3D3 to 3W	39,40	17,20	14	36		10.5		18,22
Q54	3D4 to 3W	20	17,39, 40	37	36		10.5		18,22
Q55	1D1 to 1Y	32	9,12, 13	23	21		16.5		18,21
Q56	1D5 to 1Y	12,32	9,13	5	21				
Q57	2D1 to 2Y	32	9,12, 13	7	8				
Q58	2D5 to 2Y	12,32	9,13	30	8				
Q59	3D1 to 3Y	39	17,20, 40	15	16				
Q60	3D5 to 3Y	20,39	17,40	18	16		16.5		18,21
Q61	1-2A to 1W	2	9,12, 13,23	32	22		18.0		18,20
Q62	1-2B to 1W	5,12, 32	4,9	13	22				
Q63	1-2C to 1W	13,24, 32	4,9	12	22				
Q64	1-2A to 2W Propagation delay time (t_{PLH}), C_L = 150 pF	28	7,9, 12,13	32	29	4.0	18.0	ns	18,20

Source: https://assist.dla.mil -- Downloaded: 202640-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q65	1-2B to 2W Propagation delay time (t_{PLH}), C_L = 150 pF	12,30, 32	9,11	13	29	4.0	18.0	ns	18,20
Q66	1-2C to 2W	6,13, 32	9,11	12	29				
Q67	3A to 3W	35	15,17, 20,40	39	36				
Q68	3B to 3W	18,20, 39	17,19	40	36				
Q69	3C to 3W	14,39, 40	17,19	20	36		18.0		18,20
Q70	1-2A to 1Y	23	2,9, 12,13	32	21		25.0		18,21
Q71	1-2B to 2Y	11,12, 32	9,30	13	8		25.0		18,21
Q72	3C to 3Y	19,39, 40	14,17	20	16		25.0		18,21
Q73	STROBE 1-2 to 1W	2	12,13, 32	9	22		16.0		18,20
Q74	STROBE 1-2 to 2W	28	12,13, 32	9	29		16.0		18,20
Q75	STROBE 3 to 3W	35	20,39, 40	17	36		16.0		18,20
Q76	STROBE 1-2 to 1Y Propagation delay time (t_{PLH}), C_L = 150 pF	2	12,13, 32	9	21	4.0	22.5	ns	18,22

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q77	$\overline{STROBE}$ 1-2 to 2Y Propagation delay time (t_{PLH}), C_L = 150 pF	28	12,13, 32	9	8	4.0	22.5	ns	18,22
Q78	$\overline{STROBE}$ 3 to 3Y Propagation delay time (t_{PLH}), C_L = 150 pF	35	20,39, 40	17	16	4.0	22.5		
R1	1D0 to 1W Propagation delay time (t_{PHL}), C_L = 50 pF	Open	9,12, 13,32	2	22	2.0	8.5		
R2	1D4 to 1W	12	9,13, 32	26					
R3	1D5 to 1W	12,32	9,13	5					
R4	1D6 to 1W	12,13	9,32	25					
R5	1D7 to 1W	12,13, 32	9	4	22				
R6	2D0 to 2W	Open	9,12, 13,32	28	29				
R7	2D4 to 2W	12	9,13, 32	33					
R8	2D5 to 2W	12,32	9,13	30					
R9	2D6 to 2W	12,13	9,32	31					
R10	2D7 to 2W	12,13, 32	9	11	29				
R11	3D0 to 3W Propagation delay time (t_{PHL}), C_L = 50 pF	Open	17,20, 39,40	35	36	2.0	8.5	ns	18,22

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R12	3D4 to 3W Propagation delay time (t_{PHL}), C_L = 50 pF	20	17,39, 40	37	36	2.0	8.5	ns	18,22
R13	3D5 to 3W	20,39	17,40	18	↓	↓	↓	↓	↓
R14	3D6 to 3W	20,40	17,39	38	↓	↓	↓	↓	↓
R15	3D7 to 3W	20,39, 40	17	19	36	↓	8.5	↓	18,22
R16	1D1 to 1Y	32	9,12, 13	23	21	↓	13.5	↓	18,21
R17	1D5 to 1Y	12,32	9,13	5	21	↓	↓	↓	↓
R18	2D1 to 2Y	32	9,12, 13	7	8	↓	↓	↓	↓
R19	2D5 to 2Y	12,32	9,13	30	8	↓	↓	↓	↓
R20	3D1 to 3Y	39	17,20, 40	15	16	↓	↓	↓	↓
R21	3D5 to 3Y	20,39	17,40	18	16	↓	13.5	↓	18,21
R22	1-2A to 1W	2	9,12, 13,23	32	22	↓	16.5	↓	18,20
R23	1-2B to 1W	5,12, 32	4,9	13	22	↓	↓	↓	↓
R24	1-2C to 1W	13,24, 32	4,9	12	22	↓	↓	↓	↓
R25	1-2A to 2W Propagation delay time (t_{PHL}), C_L = 50 pF	28	7,9 12,13	32	29	2.0	16.5	ns	18,20

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±3°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R26	1-2B to 2W Propagation delay time (t_{PHL}), C_L = 50 pF	12,30, 32	9,11	13	29	2.0	16.5	ns	18,20
R27	1-2C to 2W	6,13, 32	9,11	12	29				
R28	3A to 3W	35	15,17, 20,40	39	36				
R29	3B to 3W	18,20, 39	17,19	40	36				
R30	3C to 3W	14,39, 40	17,19	20	36		16.5		18,20
R31	1-2A to 1Y	23	2,9, 12,13	32	21		23.0		18,21
R32	1-2B to 2Y	11,12, 32	9,30	13	8		23.0		18,21
R33	3C to 3Y	19,39, 40	14,17	20	16		23.0		18,21
R34	STROBE 1-2 to 1W	2	12,13, 32	9	22		14.0		18,20
R35	STROBE 1-2 to 2W	28	12,13, 32	9	29		14.0		18,20
R36	STROBE 3 to 3W	35	20,39, 40	17	36		14.0		18,20
R37	STROBE 1-2 to 1Y Propagation delay time (t_{PHL}), C_L = 50 pF	2	12,13, 32	9	21	2.0	20.5	ns	18,22

Source: https://assist.dla.mil -- Downloaded: 20260-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R38	STROBE 1-2 to 2Y	Propagation delay time (t_{PHL}), C_L = 50 pF	28	12,13, 32	9	8	2.0	20.5	ns	18,22
R39	STROBE 3 to 3Y	Propagation delay time (t_{PHL}), C_L = 50 pF	35	20,39, 40	17	16	2.0	20.5		
R40	1D0 to 1W	Propagation delay time (t_{PHL}), C_L = 150 pF	Open	9,12, 13,32	2	22	4.0	10.5		
R41	1D4 to 1W		12	9,13, 32	26					
R42	1D5 to 1W		12,32	9,13	5					
R43	1D6 to 1W		12,13	9,32	25					
R44	1D7 to 1W		12,13, 32	9	4	22				
R45	2D0 to 2W		Open	9,12, 13,32	28	29				
R46	2D4 to 2W		12	9,13, 32	33					
R47	2D5 to 2W		12,32	9,13	30					
R48	2D6 to 2W		12,13	9,32	31					
R49	2D7 to 2W		12,13, 32	9	11	29				
R50	3D0 to 3W	Propagation delay time (t_{PHL}), C_L = 150 pF	Open	17,20, 39,40	35	36	4.0	10.5	ns	18,22

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R51	3D4 to 3W Propagation delay time (t_{PHL}), C_L = 150 pF	20	17,39, 40	37	36	4.0	10.5	ns	18,22
R52	3D5 to 3W	20,39	17,40	18					
R53	3D6 to 3W	20,40	17,39	38					
R54	3D7 to 3W	20,39, 40	17	19	36		10.5		18,22
R55	1D1 to 1Y	32	9,12, 13	23	21		15.5		18,21
R56	1D5 to 1Y	12,32	9,13	5	21				
R57	2D1 to 2Y	32	9,12, 13	7	8				
R58	2D5 to 2Y	12,32	9,13	30	8				
R59	3D1 to 3Y	39	17,20, 40	15	16				
R60	3D5 to 3Y	20,39	17,40	18	16		15.5		18,21
R61	1-2A to 1W	2	9,12, 13,23	32	22		18.0		18,20
R62	1-2B to 1W	5,12, 32	4,9	13	22				
R63	1-2C to 1W	13,24, 32	4,9	12	22				
R64	1-2A to 2W Propagation delay time (t_{PHL}), C_L = 150 pF	28	7,9, 12,13	32	29	4.0	18.0	ns	18,20

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

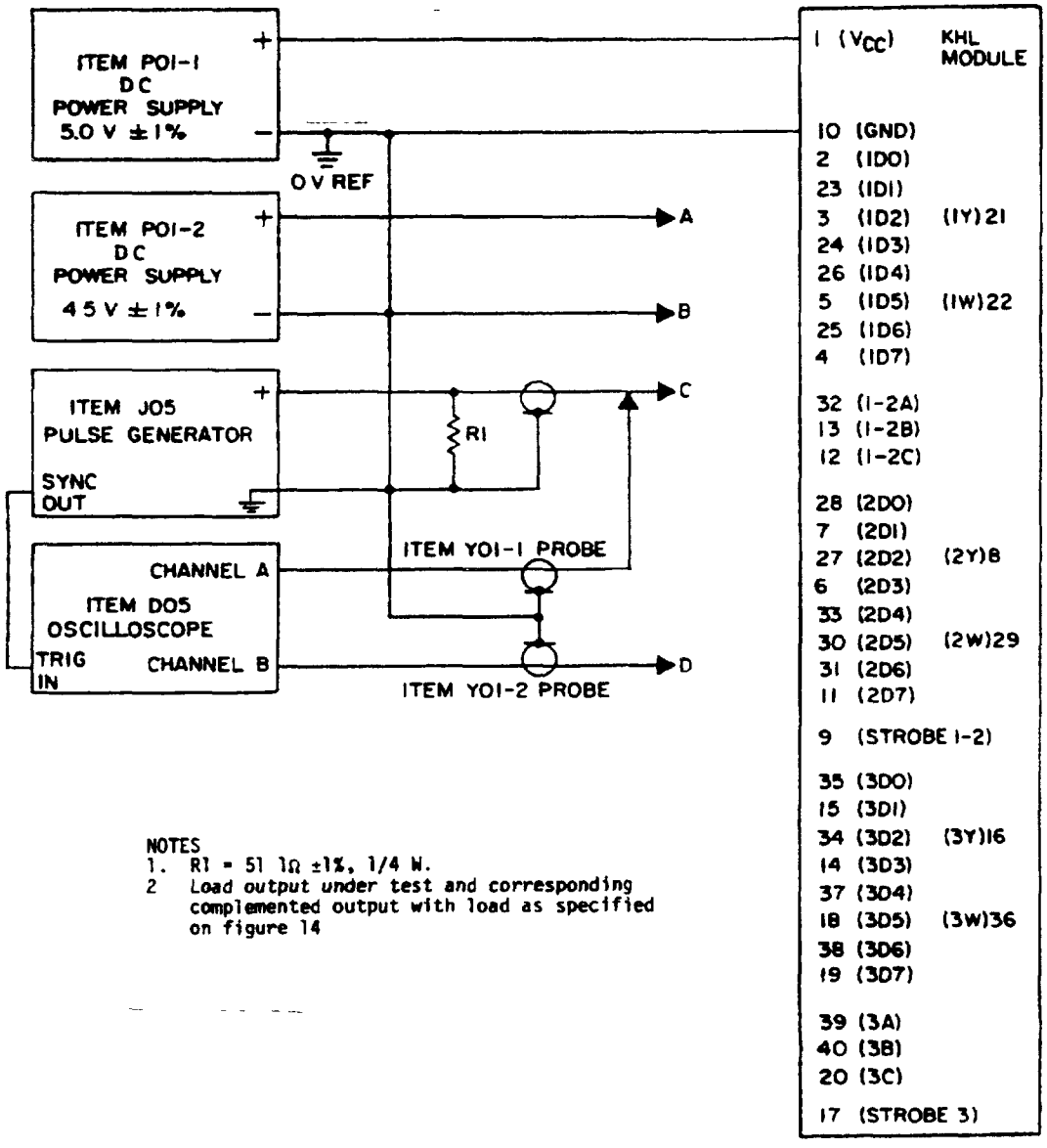
Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R65	1-2B to 2W Propagation delay time (t_{PHL}), C_L = 150 pF	12,30, 32	9,11	13	29	4.0	18.0	ns	18,20
R66	1-2C to 2W	6,13, 32	9,11	12	29				
R67	3A to 3W	35	15,17, 20,40	39	36				
R68	3B to 3W	18,20, 39	17,19	40	36				
R69	3C to 3W	14,39, 40	17,19	20	36		18.0		18,20
R70	1-2A to 1Y	23	2,9, 12,13	32	21		25.0		18,21
R71	1-2B to 2Y	11,12, 32	9,30	13	8		25.0		18,21
R72	3C to 3Y	19,39, 40	14,17	20	16		25.0		18,21
R73	STROBE 1-2 to 1W	2	12,13, 32	9	22		16.0		18,20
R74	STROBE 1-2 to 2W	28	12,13, 32	9	29		16.0		18,20
R75	STROBE 3 to 3W	35	20,39, 40	17	36		16.0		18,20
R76	STROBE 1-2 to 1Y Propagation delay time (t_{PHL}), C_L = 150 pF	2	12,13, 32	9	21	4.0	22.5	ns	18,22

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R77	STROBE 1-2 to 2Y Propagation delay time (t_{PHL}), C_L = 150 pF	28	12,13, 32	9	8	4.0	22.5	ns	18,22
R78	STROBE 3 to 3Y Propagation delay time (t_{PHL}), C_L = 150 pF	35	20,39, 40	17	16	4.0	22.5	ns	18,22

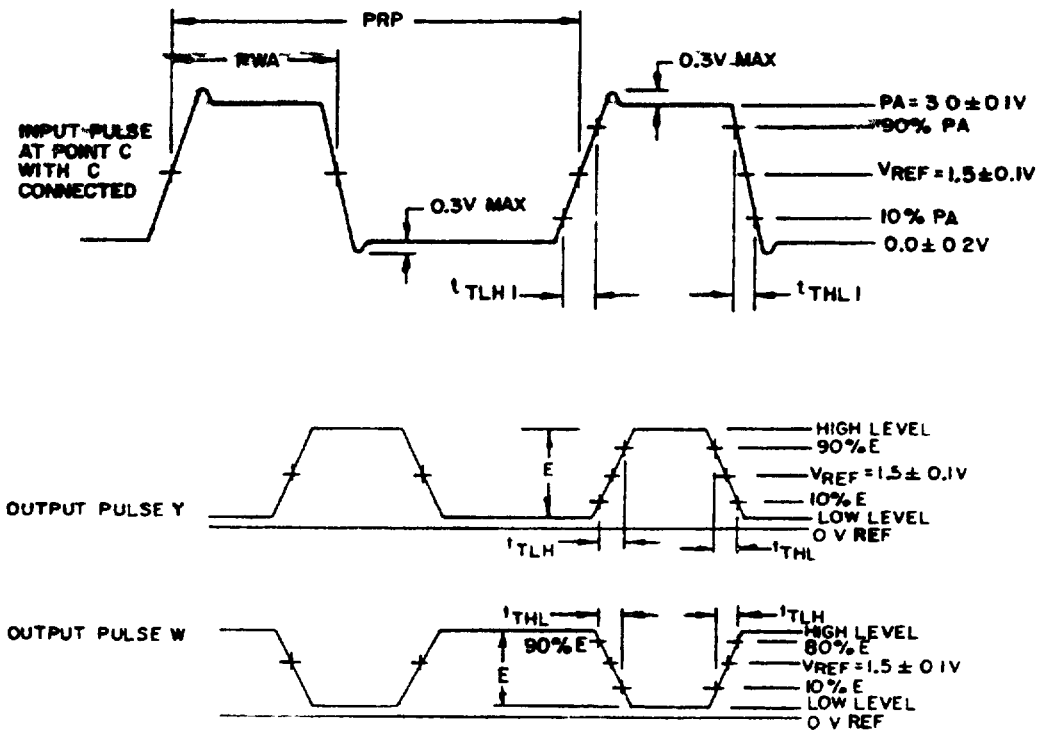
Source: https://assist.dla.mil -- Downloaded: 202640-20T15:59Z
Check the source to verify that this is the current version before use.



NOTES
1. R1 = 51 1Ω ±1%, 1/4 W.
2 Load output under test and corresponding complemented output with load as specified on figure 14

FIGURE 18 Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XV.

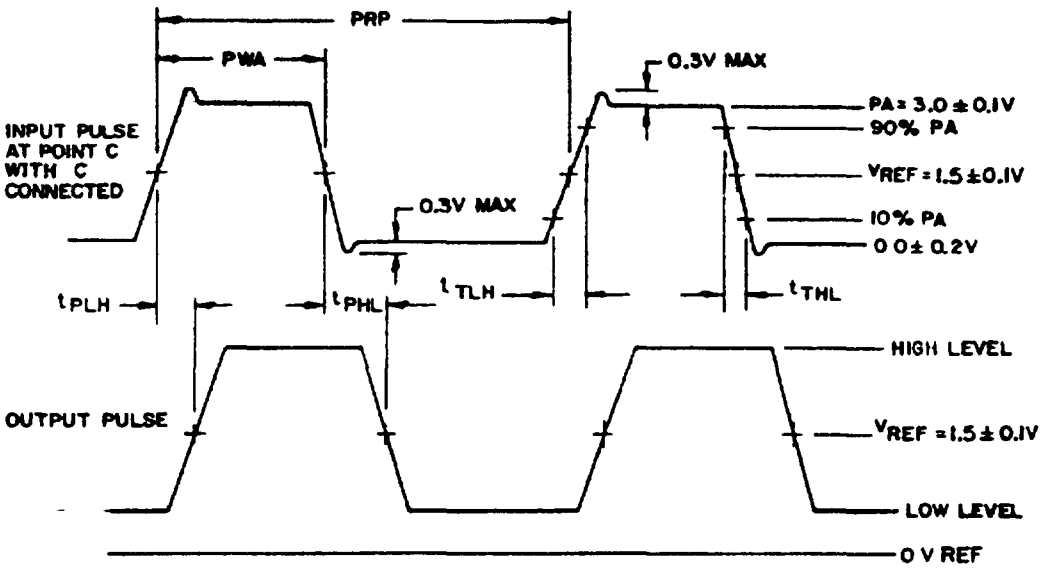
Source: https://assist.dla.mil -- Downloaded: 202610-20T15:59Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters:
   $t_{TLH1} = t_{THL1} = 4 \pm 1$ ns.
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. Before making the measurements specified in table XV, verify that the output pulse high level is $\geq$ 2.5 V and the low level is $\leq$ 0.5 V.
3. The lead length of electrical conductors between terminator resistor R1 and input pins, probes YO1-1, YO1-2 to output pins and between load circuits and output pins shall be less than 1 inch (25.4 mm)
4. t_{TLH} is measured from the 10 to 80% levels for test conditions N2, N4, N6, N8, N10 and N12 only.
5. All voltages are referenced to ground.

FIGURE 19 Transition time (t_{TLH} and t_{THL}) test circuit waveforms for table XV

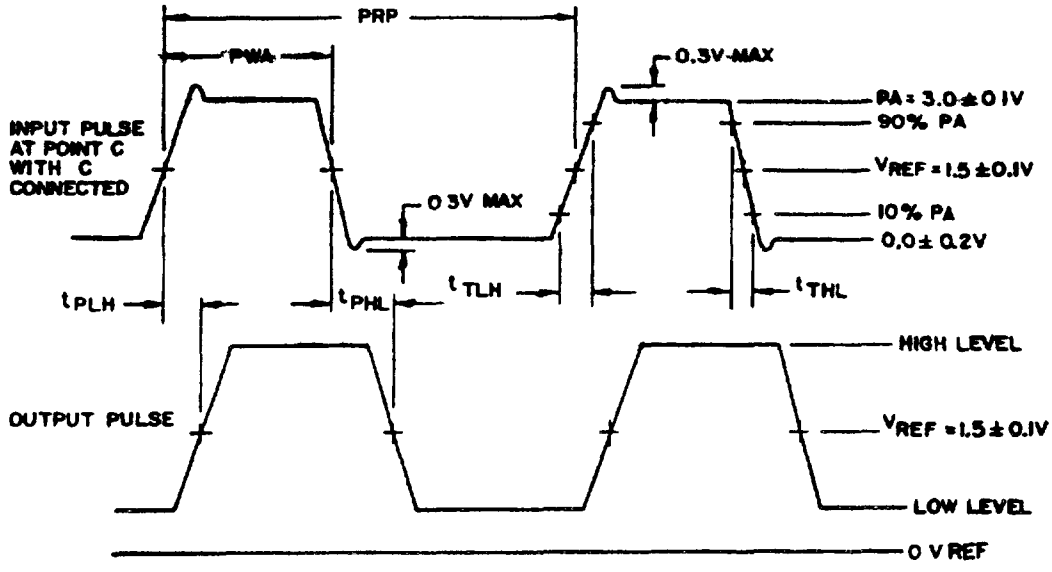
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:59Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters.
   $t_{TLH} = t_{THL} = 4 \pm 1$ ns.
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. Before making the measurements specified in table XV, verify that the output pulse high level is $\geq$ 2.5 V and the low level is $\leq$ 0.5 V.
3. The lead length of electrical conductors between terminator resistor R1 and input pins, probes YO1-1, YO1-2 to output pins and between load circuits and output pins shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

FIGURE 20. Propagation delay time (select and STROBE to W output) test circuit waveforms for table XV.

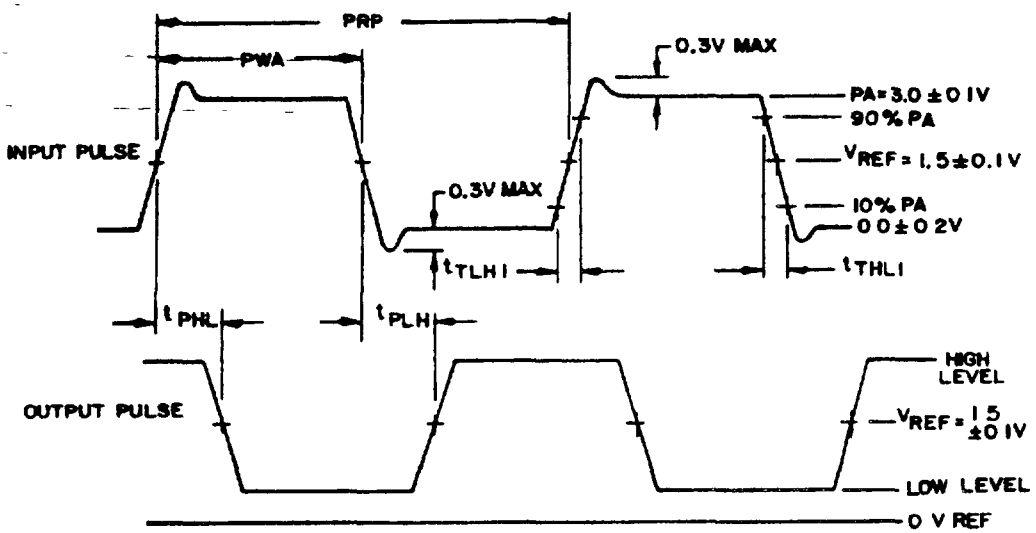
Source: https://assist.dla.mil -- Downloaded: 202610-20T15:59Z
Check the source to verify that this is the current version before use.



NOTES.
1. Input pulse parameters.
   $t_{TLH} = t_{THL} = 4 \pm 1$ ns.
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. Before making the measurements specified in table XV, verify that the output pulse high level is $\geq$ 2.5 V and the low level is $\leq$ 0.5 V.
3. The lead length of electrical conductors between terminator resistor R1 and input pins, probes Y01-1, Y01-2 to output pins and between load circuits and output pins shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

FIGURE 21. Propagation delay time (data and select to Y output) test circuit waveforms for table XV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:59Z
Check the source to verify that this is the current version before use.



NOTES.
1. Input pulse parameters
   $t_{TLH1} = t_{THL1} = 4 \pm 1$ ns.
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. Before making the measurements specified in table XV, verify that the output pulse high level is $\geq$ 2.5 V and the low level is $\leq$ 0.5 V.
3. The lead length of electrical conductors between terminator resistor R1 and input pins, probes YO1-1, YO1-2 to output pins and between load circuits and output pins shall be less than 1 inch (25.4 mm).
4 All voltages are referenced to ground.

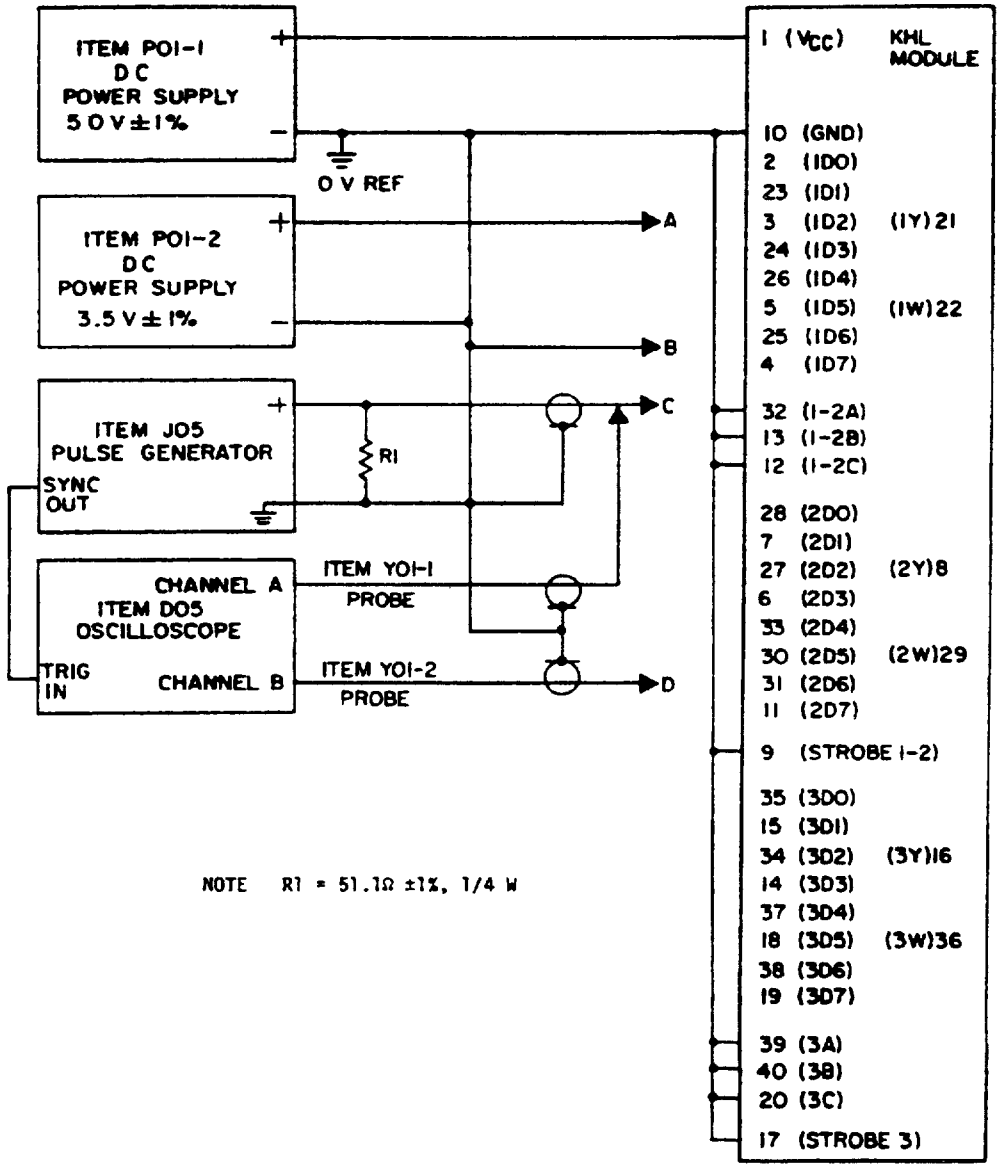
FIGURE 22. Propagation delay time (STROBE to Y output, data to W) test circuit waveforms for table XV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE XVI. Output crosstalk (low-level and high-level) test

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S1	1Y Output crosstalk, low level	Open	2	28,35	21	0	500	mV	23,24
S2	1W	2	Open	28,35	22				
S3	2Y	Open	28	2,35	8				
S4	2W	28	Open	2,35	29				
S5	3Y	Open	35	2,28	16				
S6	3W Output crosstalk, low level	35	Open	2,28	36		500		
T1	1Y Output crosstalk, high level	2	Open	28,35	21		400		
T2	1W	Open	2	28,35	22				
T3	2Y	28	Open	2,35	8				
T4	2W	Open	28	2,35	29				
T5	3Y	35	Open	2,28	16				
T6	3W Output crosstalk, high level	Open	35	2,28	36	0	400	mV	23,24

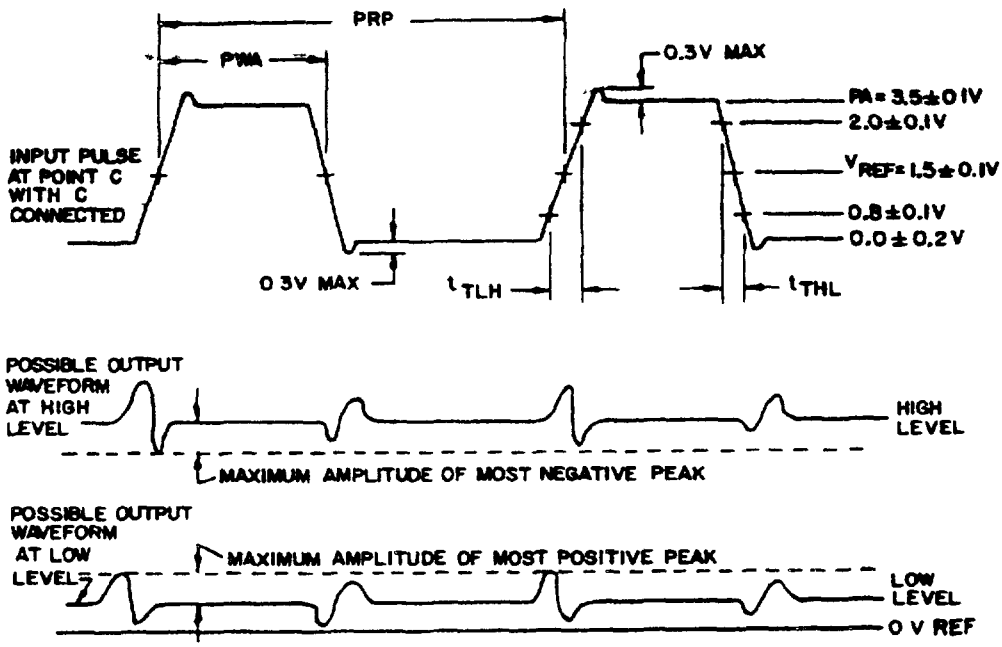
Source: https://assist.dla.mil -- Downloaded: 2026-02-20T15:59Z
Check the source to verify that this is the current version before use.



NOTE R1 = 51.1Ω ±1%, 1/4 W

FIGURE 23. Output crosstalk (low-level and high-level) test circuit for table XVI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:59Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters:
   $t_{THL} = t_{TLH} = 3.5 \pm 1.5$ ns (verified at each input pin).
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. The lead length of electrical conductors between terminator resistor R1 and input pins, probes YO1-1, YO1-2 to input and output pins, load circuits to output pins, and ground leads of probes YO1-1, YO1-2 to test fixture and module ground pins shall be less than 1 inch (25.4 mm).

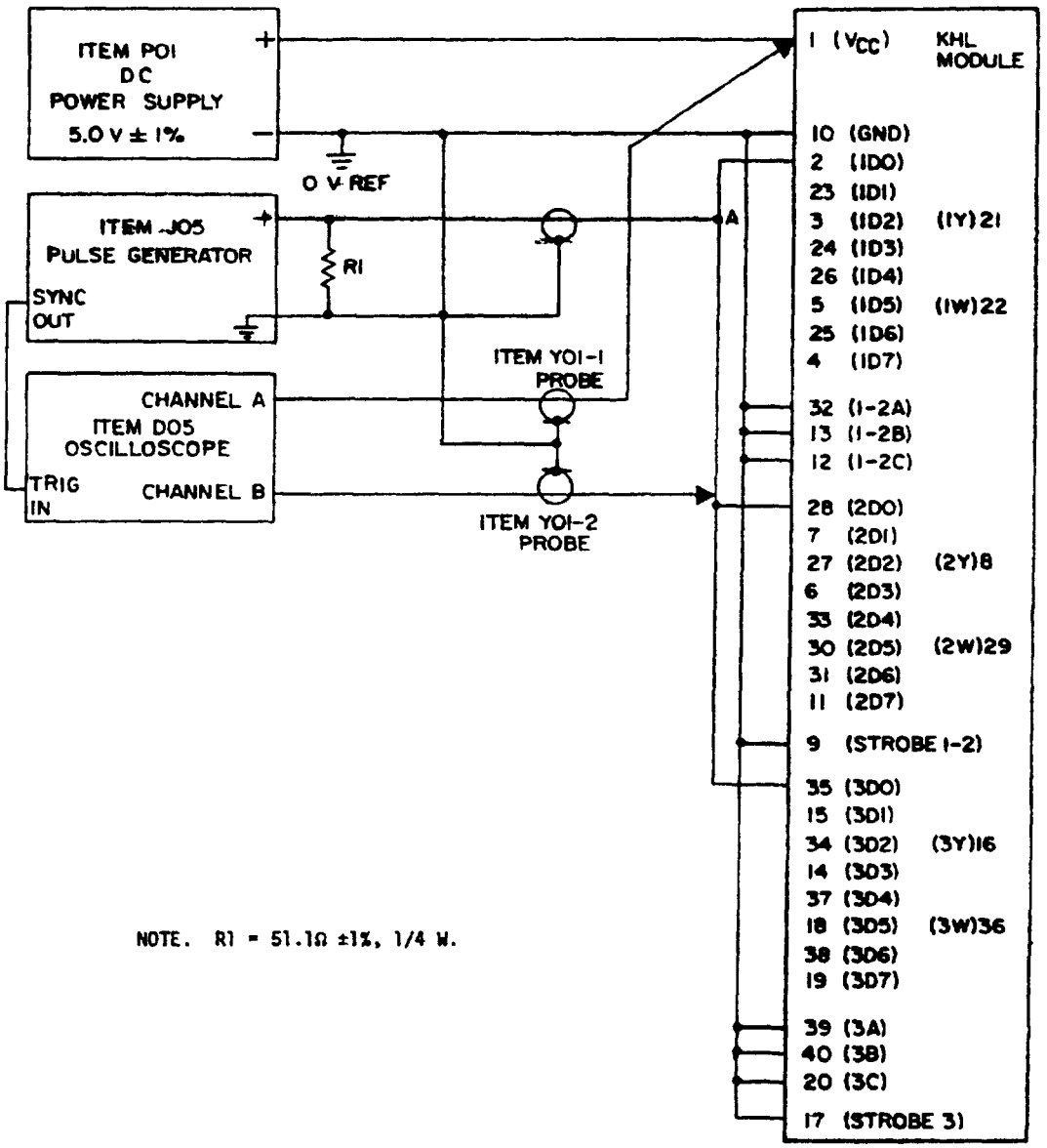
FIGURE 24 Output crosstalk tests (low-level and high-level) test circuit waveforms for table XVI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE XVII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25° +5°C Ambient		Unit	Test figure
		Min	Max		
U	Power supply transient amplitude	0	400	mV	25,26

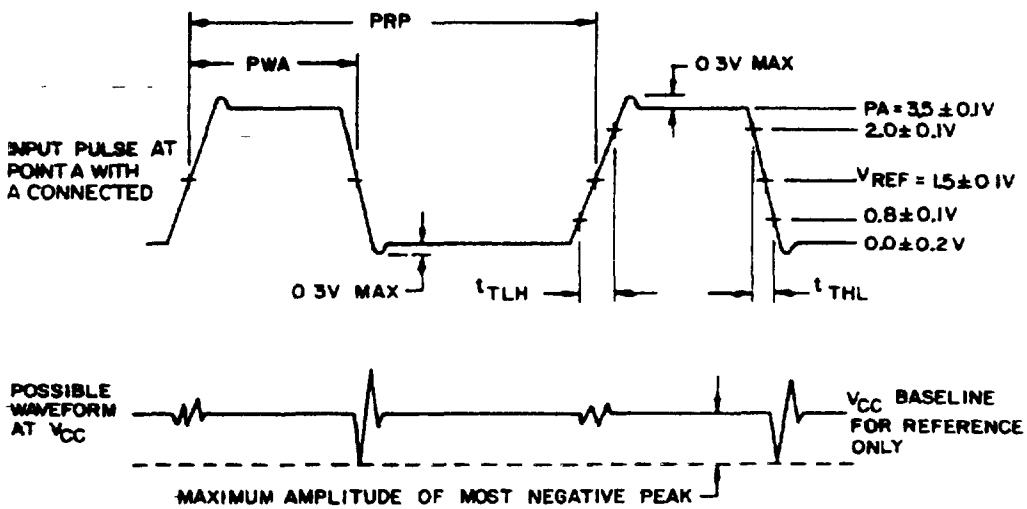
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:59Z
Check the source to verify that this is the current version before use.



NOTE. R1 = 51.1Ω ±1%, 1/4 W.

FIGURE 25 Power supply transient amplitude test circuit for table XVII

Source: https://assist.dla.mil -- Downloaded: 202640-20T15:59Z
Check the source to verify that this is the current version before use.



NOTE. Input pulse parameters
$t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin)
PWA = 500 ±50 ns.
PRP = 1,000 ±100 ns

FIGURE 26 Power supply transient amplitude test circuit waveforms for table XVII

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE XVIII. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
V	Power supply filter capacitance	Open	1	1.6	3.0	µF	27

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE XIX. Input capacitance test.

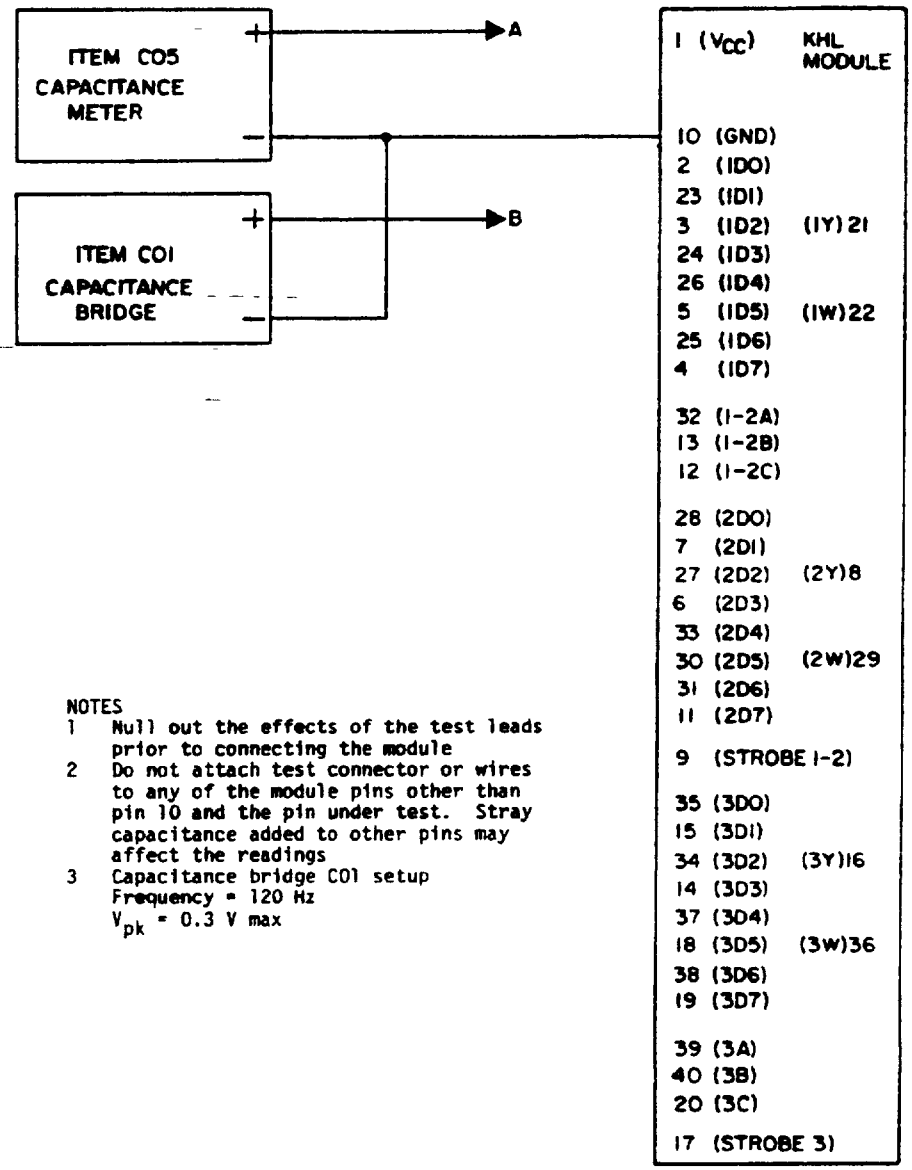
Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
W1	1D0	Input capacitance	2	Open	5.0	15.0	pF	27
W2	1D1		23					
W3	1D2		3					
W4	1D3		24					
W5	1D4		26					
W6	1D5		5					
W7	1D6		25					
W8	1D7		4					
W9	2D0		28					
W10	2D1		7					
W11	2D2		27					
W12	2D3		6					
W13	2D4		33					
W14	2D5		30					
W15	2D6		31					
W16	2D7		11					
W17	3D0		35					
W18	3D1	Input capacitance	15	Open	5.0	15.0	pF	27

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:59Z
Check the source to verify that this is the current version before use.

TABLE XIX. Input capacitance test - Continued.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
W19	3D2	Input capacitance	34	Open	5.0	15.0	pF	27
W20	3D3		14					
W21	3D4		37					
W22	3D5		18					
W23	3D6		38					
W24	3D7		19					
W25	3A		39					
W26	3B		40					
W27	3C		20					
W28	STROBE 3		17			15.0		
W29	1-2A		32			30.0		
W30	1-2B		13					
W31	1-2C		12					
W32	STROBE 1-2	Input capacitance	9	Open	5.0	30.0	pF	27

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:59Z
Check the source to verify that this is the current version before use.



NOTES
1 Null out the effects of the test leads prior to connecting the module
2 Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings
3 Capacitance bridge C01 setup
Frequency = 120 Hz
V_{pk} = 0.3 V max

FIGURE 27. Power supply filter and input capacitance test circuit for tables XVIII and XIX.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:59Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D.C. 20360

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

POSTAGE AND FEES PAID
NAVY DEPARTMENT
DOD 316

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D.C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 202640-20T15:59Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:59Z
Check the source to verify that this is the current version before use.